GATI SHAKTI VISHWAVIDYALAYA ▪ VADODARA

A Central University under the Ministry of Railways, Government of India

B.TECH. THESIS ▪ CIVIL ENGINEERING ▪ RAIL ENGINEERING

ASSESSMENT OF RAILROAD BALLAST DEGRADATION

under modified drop-weight impact loading

and image-based morphological analysis

submitted by

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under the supervision of

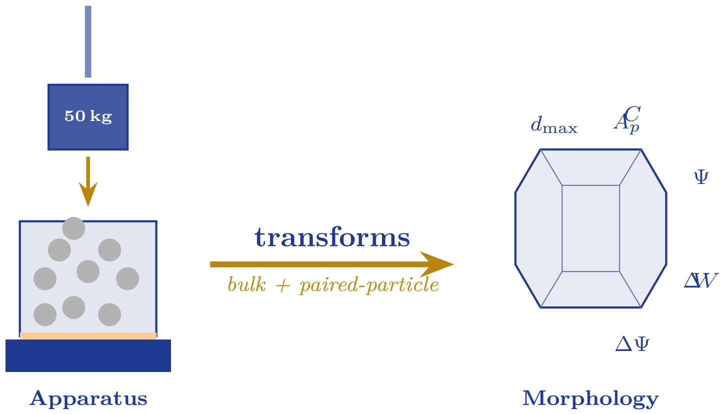
Dr. Dinesh Gundavaram

Assistant Professor, School of Civil Engineering

APRIL 2026

Lalbaug, Vadodara 390004, Gujarat, India

Frontispiece



The two-stream methodology of this thesis: large-scale impact loading on the left, paired pre-and-post 3D morphological analysis on the right. The two streams converge in the Indian Railway Fouling Index FI_{IN} *and a 22-descriptor per-particle record.*

A B.Tech. Thesis in Civil Engineering | Specialisation: Rail Engineering

ASSESSMENT OF RAILROAD BALLAST DEGRADATION UNDER MODIFIED DROP-WEIGHT IMPACT LOADING CONDITIONS AND IMAGE-BASED MORPHOLOGICAL ANALYSIS

submitted by

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In Partial Fulfilment of the Requirements for the Degree of

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A Central University under the Ministry of Railways, Government of India

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Copyright

Patent notice

The apparatus described in Chapter 3 is the subject of an Indian utility-patent application filed by Gundavaram, D. and Legha, D. (*Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions*, 2026, draft prepared).

Document version

Version 2.0, compiled May 21, 2026 with pdfLaTeX, BibTeX, and the `natbib`, `glossaries`, `cleveref`, `siunitx`, `tcolorbox`, and `TikZ` packages.

Open materials

The morphological-analysis pipeline (`ballast_analyzer.py`) is reproduced in Appendix B. The raw paired Stereolithography (triangular surface-mesh file format) (STL) meshes and the laboratory spreadsheets are archived at Gati Shakti Vishwavidyalaya and are available from the authors on reasonable request for non-commercial research use.

Typeset in Latin Modern on A4 paper.

Dedicated to

our families,

our supervisor, and

the engineers and operators of Indian Railways,

whose patience and trust made this work possible.

"Track maintenance is not the cost of running a railway; it is *the railway. The ballast is where the train meets the earth, and where the earth answers back."*

— COENRAAD ESVELD, *Modern Railway Track*

"Without data you're just another person with an opinion."

— W. EDWARDS DEMING

"We see further by standing on the shoulders of those who laid the sleepers before us."

— *after* BERNARD OF CHARTRES

Candidate's Declaration

We hereby certify that

(i) The work contained in this thesis is original and has been carried out by us jointly under the guidance of our supervisor, Dr. Dinesh Gundavaram, Gati Shakti Vishwavidyalaya, Vadodara.

(ii) The work has not been submitted in whole or in part to any other institute or university for the award of any degree, diploma, or certificate.

(iii) We have followed the academic guidelines and ethical code of conduct provided by Gati Shakti Vishwavidyalaya in preparing this thesis.

(iv) Wherever we have used materials (data, experimental results, theoretical analyses, figures, tables, or text) from other sources, we have given due credit to them by citing them in the text of the thesis and by giving full bibliographic details in the reference list. Where required, permission from the copyright owners has been obtained.

(v) The experimental programme reported in this thesis was conducted by both candidates jointly at Gati Shakti Vishwavidyalaya.

Durgesh Kumar Legha
(2270007)

Sandeep Patel
(2280043)

Certificate of Thesis Supervisor

This is to certify that the thesis entitled *"Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions and Image-Based Morphological Analysis"*, submitted by Durgesh Kumar Legha (Roll No. 2270007) and Sandeep Patel (Roll No. 2280043) to Gati Shakti Vishwavidyalaya, is a record of bonafide research work carried out jointly by them under my supervision. The work embodied in this thesis has not been submitted, in whole or in part, for the award of any degree, diploma, or certificate at any other institute or university. In my opinion, this thesis is of the standard required for the degree of Bachelor of Technology in Civil Engineering (Specialisation: Railway Engineering), and I consider it worthy of consideration for the said degree.

Dr. Dinesh Gundavaram
Thesis Supervisor
Assistant Professor
School of Civil Engineering,
Gati Shakti Vishwavidyalaya,
Vadodara, Gujarat – 390004.

Date: ____________________
Place: Vadodara

Certificate of Approval

Certified that the thesis entitled *"Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions and Image-Based Morphological Analysis"* submitted by Durgesh Kumar Legha (Roll No. 2270007) and Sandeep Patel (Roll No. 2280043) to Gati Shakti Vishwavidyalaya for the award of the Degree of Bachelor of Technology in Civil Engineering (Specialisation: Railway Engineering) has been accepted by the B. Tech. project committee members. The project presentation by the candidates was held on 23rd April 2026.

Dr. Dinesh Gundavaram
Thesis Supervisor and
B. Tech. Project Coordinator

Dr. Dinesh Gundavaram
B. Tech. Project Coordinator

Dr. Pradeep
B. Tech. Programme Coordinator

Acknowledgement

We wish to place on record our deep sense of gratitude to all those without whose guidance, encouragement, and support this work would not have reached its present form.

Our foremost and most profound thanks are due to our supervisor, Dr. Dinesh Gundavaram, Assistant Professor at Gati Shakti Vishwavidyalaya. Across every stage of this project—from the first conversation about the limitations of the conventional Aggregate Impact Value test, through the careful re-conception of the large-scale drop-weight apparatus, the design of the nine-configuration factorial matrix, the painstaking three-dimensional scanning work, and finally the writing of this thesis—he has been a patient and rigorous mentor. The intellectual standards he holds and the clarity with which he insisted on first-principles reasoning are reflected in every chapter that follows. Any errors or shortcomings that remain are our own.

We express our sincere gratitude to the Hon'ble Vice Chancellor of Gati Shakti Vishwavidyalaya for fostering an academic environment that encourages research, innovation, and critical thinking. We are also thankful to the Programme Coordinator and all faculty members for their structured guidance, timely reviews, and valuable feedback, which helped keep this work on track.

We extend our appreciation to the laboratory technicians and technical staff of Gati Shakti Vishwavidyalaya for their assistance and cooperation during the experimental phases of this project. We also acknowledge the support extended by other departments and faculty members whose indirect contributions and institutional support facilitated the smooth execution of this work.

On a personal note, we thank our families whose belief in the value of this work was never conditional on the stage of its results, and our batchmates in the Civil Engineering programme, whose conversations sharpened many of the mechanistic interpretations that appear in the discussion chapter. In particular we are grateful to the friends who kept us company through late laboratory evenings on more occasions than we care to count.

Finally, we acknowledge the broader community of railway geotechnics researchers whose published work we have cited in this thesis. The scientific literature is a cumulative enterprise, and every modest contribution in this document rests on decades of careful experimentation and thought by others.

Durgesh & Sandeep

Abstract

The thesis focuses on addressing the critical issue of railway ballast degradation, which significantly impacts the safety, stability, and operational efficiency of railway tracks. Tailored specifically to the unique conditions of Indian Railways, this research aims to develop a deeper understanding of ballast behaviour under impact loading and to predict the quantum of ballast degradation using 3D image processing as well as machine-learning models.

At the core of the study is the modification of the European-standard drop-weight impact test apparatus to better align with Indian railway ballast specifications and operational conditions. The modified setup accurately replicates on-field stress conditions experienced by ballast in the Indian context, providing a more relevant and reliable framework for analysis. The modified apparatus serves as a critical tool for evaluating ballast degradation under controlled yet realistic conditions, delivering nominal contact stresses of 275 kPa (at a 300 mm drop height) and 523 kPa (at a 450 mm drop height) — corresponding to the 22.5 t axle load characteristic of Indian Railways.

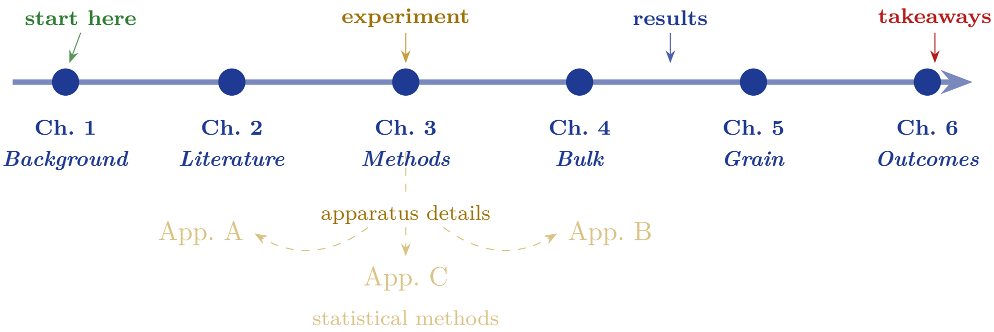
A significant innovation in this study is the use of advanced 3D image processing techniques to evaluate morphological changes in ballast particles before and after impact testing. High-resolution 3D meshes are captured to analyse changes in particle shape, size, and texture, providing a detailed and quantitative understanding of ballast degradation. A paired-particle protocol applied to 45 marked tracer particles across nine experimental configurations enables precise measurement of deterioration, allowing identification of critical thresholds and damage modes (Diffuse corner abrasion (present-study classification) (Mode A) — diffuse abrasion vs. Catastrophic single-particle bulk fracture (present-study classification) (Mode B) — catastrophic fracture) that are not detectable through traditional sieve analysis.

The outcomes of this study contribute to railroad ballast degradation assessment using image-based morphology and machine learning, delivering a scalable and accurate alternative to conventional physical testing, with substantial implications for railway safety and maintenance planning.

Keywords: Railway ballast; drop-weight impact loading; Research Designs and Standards Organisation (RDSO) gradation; Fouling Index; 3D structured-light scanning; particle morphology; under-ballast mats; ballast degradation.

Thesis Highlights

The principal contributions of the thesis can be read in a single page. Each item is anchored to the chapter where it is developed in full. A reading roadmap is provided below for examiners who wish to navigate the thesis by interest area.



Apparatus

A purpose-built large-scale drop-weight impact apparatus (300 mm mould, 50 kg hammer, 200–450 mm adjustable drop, in-line 5 t load cell) was designed, fabricated, and commissioned at Gati Shakti Vishwavidyalaya. The two drop heights bracket the Indian Railways field stress envelope at 275 kPa / 147 J and 523 kPa / 221 J. (Chapter 3)

Experimental campaign

A nine-configuration factorial test matrix crossing three pad conditions (Unreinforced (no pad) (UR) / Rubber Under-Ballast Mat (12 mm) (RUBM) / Polyurethane Under-Ballast Mat (Getzner) (PUBM)) with two base conditions (Flexible Base (100 mm compacted sand layer) (FB) / Rigid Base (concrete mould floor) (RB)) and two impact energies (300 mm / 450 mm drop) was executed; the three FB–450 mm cells were excluded by design. (Chapter 3)

Bulk fouling response

The Indian Railway Fouling Index FI_{IN} spans a 6.4-fold range across the nine configurations, from 0.533 % at T1 (PUBM–FB–300) to 3.386 % at T7 (UR–RB–450). *Pad performance is energy- and base-dependent*: PUBM dominates at both extremes, but RUBM edges ahead on the rigid base at low energy. (Chapter 4)

Two damage modes (Mode A / Mode B)

A paired per-particle 3D morphological protocol on 45 tracer particles revealed two distinct damage modes. Mode A (diffuse corner abrasion) is universal: every particle exhibits a small but statistically significant sphericity rise ($\Delta\Psi = +0.0041$, $p < 10^{-6}$). Mode B (catastrophic fracture) is rare (6/45) and *occurs only on rigid-base configurations.* (Chapter 5)

Settlement vs. fouling are non-interchangeable

The test with the largest settlement (`T5`, 48.6 mm) is not the test with the largest fouling (`T7`). Settlement reflects rearrangement; fouling reflects breakage. Both metrics are required for a complete bulk-level characterisation. (Chapter 4)

Output

A utility-patent draft (Appendix B); two journal manuscripts under review; two accepted Springer conference papers; an open-source Python pipeline (`ballast_analyzer.py`) that ingests paired pre/post STL meshes and exports twenty-two morphological descriptors per particle. (Chapters 6 & B)

Lay Summary

RAILWAY tracks rest on a layer of crushed stone called *ballast*. The ballast keeps the rails in place, spreads the wheel loads to the soil beneath, and lets rainwater drain away. Every time a train passes, the ballast is pushed and bumped, and the stones slowly break down. Once the ballast becomes too crushed, the track sinks, the ride becomes rough, and the railway must spend money and time on maintenance.

This thesis asks a simple question: *how quickly does ballast break down under the conditions seen on Indian Railways, and what can we do to slow it down?* The answer matters because the Indian Railways network is the largest passenger railway in the world; even a small improvement in ballast life saves substantial maintenance cost and keeps trains running on time.

To answer the question, the authors built a heavy laboratory machine that drops a 50-kilogram hammer onto a tub full of ballast, reproducing the kind of impact that a train wheel gives to a rough section of track. By placing different rubber mats and different cushions underneath the ballast, the machine can show which combinations protect the stones best. By scanning forty-five individual ballast stones in three dimensions both *before* and *after* they are pounded, the authors can measure—for the first time on Indian ballast—exactly how the shape of each stone changes.

The thesis reports two main discoveries. First, polyurethane rubber mats placed under the ballast reduce the amount of breakage significantly, especially when the base of the track is rigid (as it is over bridges and culverts). Second, ballast stones break in *two distinct ways*: most stones lose their sharp corners gradually (a slow, manageable process), but a small fraction *shatter* catastrophically—and the shattering happens only on rigid bases. This second finding suggests that the hardest-to-maintain stretches of Indian Railways are the bridges and the transition zones at their ends, and that under-ballast mats should be deployed there as a priority.

Statement of Originality

We, **Durgesh Kumar Legha** (Roll No. 2270007) and **Sandeep Patel** (Roll No. 2280043), declare that this thesis titled *Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions and Image-Based Morphological Analysis* is our own original work, conducted between July 2024 and April 2026 at the School of Civil Engineering, Gati Shakti Vishwavidyalaya, Vadodara, under the supervision of Dr. Dinesh Gundavaram.

We further declare that:

1. All published and unpublished sources used in this thesis are cited in the text and listed in the References. The BibTeX-based bibliography contains every reference cited; there are no uncited references and no unreferenced citations.
2. All laboratory data, including the nine raw sieve-analysis spreadsheets, the in-line load-cell records, and the ninety paired STL meshes (pre- and post-test, for the forty-five tracer particles), were generated by the authors using the apparatus described in Chapter 3. No data have been fabricated, falsified, or selectively omitted.
3. The novel contributions of this thesis (the modified drop-weight apparatus, the per-particle paired protocol, the Mode-A / Mode-B damage taxonomy, and the `ballast_analyzer.py` pipeline) are, to the best of our knowledge, original to this work. Closely related prior work is critically reviewed in Chapter 2.
4. No part of this thesis has been submitted previously, either in whole or in part, for any degree at any university.
5. The two journal manuscripts under review and the two accepted Springer conference papers listed in *Outcomes of This Project* are derived from the work reported here; the relevant chapters cite those manuscripts where overlap occurs.

Durgesh Kumar Legha	**Sandeep Patel**
Roll No. 2270007	Roll No. 2280043

Vadodara, April 2026.

Statement of Contributions

The work reported in this thesis was jointly carried out by the two candidates under the supervision of Dr. Dinesh Gundavaram. The following contribution matrix follows the CRediT (Contributor Roles Taxonomy) of the National Information Standards Organization, which is increasingly adopted by international journals to attribute author contributions transparently.

CRediT role	D. K. Legha	S. Patel	D. Gundavaram
Conceptualisation	supporting	supporting	lead
Methodology	lead	lead	supporting
Apparatus design & fabrication	lead	equal	supporting
Software (`ballast_analyzer.py`)	equal	lead	supporting
Validation	equal	equal	supporting
Formal analysis	lead	equal	supporting
Investigation (experiments)	equal	equal	supporting
Resources / funding	—	—	lead
Data curation	equal	lead	supporting
Writing – original draft	lead	equal	supporting
Writing – review & editing	equal	equal	lead
Visualisation (TikZ, plots)	lead	equal	supporting
Supervision	—	—	lead
Project administration	supporting	supporting	lead

Where two candidates are marked *lead* or *equal* for the same role, the work was performed jointly and is not separable into individual products. Where one candidate is marked *lead*, that candidate took primary responsibility for the role and is the principal defender of that aspect of the work.

Chapter-by-chapter contributions. Chapters 1 and 2 were drafted by D. K. Legha and revised jointly. The apparatus design and instrumentation documented in Chapter 3 are due principally to D. K. Legha (mechanical design, drop tower) and S. Patel (load-cell integration, instrumentation, structured-light scanning protocol). Chapter 4 (bulk results) was drafted by D. K. Legha; Chapter 5 (per-particle morphology) was drafted by S. Patel. Chapter 6 was drafted jointly. The `ballast_analyzer.py` pipeline reproduced in Appendix B was written principally by S. Patel with code review and unit-test design by D. K. Legha.

Reproducibility & Data-Availability Statement

This thesis adheres to the reproducibility principles of the **FAIR Guiding Principles** (Findable, Accessible, Interoperable, Reusable) and reports its methods in sufficient detail that an independent laboratory should be able to reproduce both the apparatus and the analysis pipeline.

Apparatus All structural drawings, instrumentation diagrams, and calibration certificates for the modified drop-weight impact apparatus are archived at the School of Civil Engineering, Gati Shakti Vishwavidyalaya, Vadodara. The apparatus is the subject of an Indian utility-patent application; the structural geometry and the calibration procedure are reproduced in Chapter 3 in sufficient detail that the apparatus can be replicated.

Materials The basalt ballast is sourced from a single quarry near Vadodara; the gradation conforms to the Indian Railways RDSO specification for 65 mm-nominal ballast. The two under-ballast mats (PUBM and RUBM) are commercially available and identified by manufacturer in Sections 3.2.2 and 3.2.3. The flexible-base sand cushion and rigid-base steel plate are off-the-shelf laboratory items.

Raw data The complete laboratory record comprises:

1. nine pre-test sieve-analysis spreadsheets and nine post-test sieve-analysis spreadsheets;
2. nine in-line load-cell time-series files (one per test);
3. ninety paired STL meshes (45×2) from the EinScan SL structured-light scanner;
4. the master experimental log (Excel) cross-referencing the systematic test labels `T1`–`T9` to the legacy laboratory labels (LH, HH).

All raw data are archived on the GSV Civil Engineering server and on a redundant offline backup. The data are available from the authors on reasonable request for non-commercial research use, in accordance with the GSV data-sharing policy.

Software The morphological-analysis pipeline (`ballast_analyzer.py`, Python 3.10+, MIT licence) is reproduced in full in Appendix B and depends on the publicly available `trimesh`, `numpy`, and `scipy` libraries. The pipeline has been unit-tested against synthetic reference meshes (sphere, ellipsoid, cube) for which the descriptors have known closed-form values; the test suite is included in the script docstring.

Analysis pipeline

All statistical tests reported in Chapter 5 (Wilcoxon signed-rank test, Shapiro–Wilk normality test, Tukey 1.5×Inter-Quartile Range (IQR) outlier-detection criterion) are reproducible from the raw paired meshes using the pipeline. Random seeds are not used; the analysis is deterministic.

Compilation

The thesis itself is compiled from the LaTeX source on the GSV Civil Engineering server with the command chain `pdflatex` → `bibtex` → `makeglossaries` → `pdflatex` → `pdflatex` (or equivalently `latexmk -pdf -bibtex`). The compile is deterministic when `SOURCE_DATE_EPOCH` is set.

Dedicated to

our families,

our supervisor, and

the engineers and operators of Indian Railways,

whose patience and trust made this work possible.

Outcomes of This Project

Journal Manuscripts

1. Legha D. and Gundavaram D., *Composite Railway Sleepers: A Review of Mechanical Performance, Durability and Adoption Challenges*, *Railway Engineering Science* (2026) **(Under Review)**. (IF = 5.4)
2. Legha Durgesh K., Gundavaram Dinesh, Patel Sandeep, *A large-scale drop-weight impact testing apparatus for assessment of Indian Railway ballast: influence of stress level and subgrade–interlayer configuration on settlement and particle degradation*, *Construction and Building Materials* (2026) **(Under Review)**. (IF = 7.2)

Patents

1. Gundavaram D., Legha D., (2026), *Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions.* (Indian Utility Patent) **(Draft Prepared)**.

Conference Papers

1. Legha D.K., Patel S., Gundavaram D., (2026) *Optimization of Bitumen Content in Bitumen-Stabilized Ballast Using Flowability Characteristics*, Recent Research on Environmental Earth Sciences, MedGU 2025, Springer. **(Accepted, In Press)**.
2. Legha D., Gundavaram D., Mishra G., (2026) *Global Practices of Railway Drainage Systems with Emphasis on India and Emerging Innovation*, Geotechnical Practices for Innovations and Sustainability, Proceedings of the Indian Geotechnical Conference (IGC 2025), Lecture Notes in Civil Engineering, Springer. **(Accepted, In Press)**.

Scan the QR code for the project video

A short documentary of the apparatus design, test campaign, and analysis pipeline.

Contents

List of Figures

List of Tables

List of Abbreviations

AAV	Aggregate Abrasion Value
AIV	Aggregate Impact Value
ANOVA	Analysis of Variance
BIS	Bureau of Indian Standards
BTP	B.Tech. Project
CAD	Computer-Aided Design
DEM	Discrete Element Method
EN	European Norm / European Standard
FB	Flexible Base (100 mm compacted sand layer)
FI	Fouling Index (after Selig & Waters, 1994)
FI_{IN}	Indian Railway Fouling Index
GSV	Gati Shakti Vishwavidyalaya
HH	High Height (450 mm drop) — legacy laboratory label
IPO	Indian Patent Office
IQR	Inter-Quartile Range
IRJ	Insulated Rail Joint
IS	Indian Standard
LAA	Los Angeles Abrasion
LH	Low Height (300 mm drop) — legacy laboratory label
Mode A	Diffuse corner abrasion (present-study classification)
Mode B	Catastrophic single-particle bulk fracture (present-study classification)
PSD	Particle Size Distribution
PUBM	Polyurethane Under-Ballast Mat (Getzner)
RB	Rigid Base (concrete mould floor)
RDSO	Research Designs and Standards Organisation

RUBM	Rubber Under-Ballast Mat (12 mm)
SciPy	Scientific Python library
SEM	Scanning Electron Microscopy
SL	Structured-Light (scanning)
STL	Stereolithography (triangular surface-mesh file format)
UBM	Under-Ballast Mat
UIC	Union Internationale des Chemins de Fer
UR	Unreinforced (no pad)
USP	Under-Sleeper Pad

Glossary of Terms

RDSO gradation
: The particle-size envelope specified by Indian Railways for railway ballast: 100 % passing the 65 mm sieve and 0 % passing the 20 mm sieve, with intermediate retention bands specified at 40 mm and 25 mm.

ballast
: The layer of crushed, angular stone (typically 20–65 mm) placed beneath and around the sleepers of a railway track. It distributes wheel loads to the subgrade, restrains lateral and longitudinal track movement, and provides rapid drainage.

convexity
: A dimensionless particle-shape descriptor measuring the fraction of the particle's convex-hull volume occupied by the particle itself. Defined as $C = V/V_{\mathrm{ch}}$. Bounded above by unity (a strictly convex particle).

fines
: Particles passing the 0.075 mm sieve. Produced in ballast by attrition (corner-by-corner abrasion) and, more dramatically, by single-particle fracture (Mode B). Accumulation of fines is the dominant driver of ballast fouling.

fouling
: The accumulation of fines (<0.075 mm) and small fragments within the ballast voids. Caused by breakage of ballast particles under repeated loading, infiltration of subgrade fines, atmospheric deposition, or spillage of cargo. Reduces drainage and inter-particle friction.

paired (per-particle) protocol
: An experimental design in which the *same* physical specimen is measured before and after a treatment, and the change is computed on a per-specimen basis. Reduces between-specimen variability and is the appropriate test when natural particle-to-particle scatter is large compared to the expected effect.

shake-down regime
: The post-densification steady state in which a granular layer responds to additional cyclic loading with small, near-elastic increments of permanent strain rather than with continued plastic densification. Reaching the shake-down regime is the conventional endpoint for laboratory cyclic-loading tests.

sphericity
: A dimensionless particle-shape descriptor measuring how closely a particle approaches a perfect sphere. Defined as $\Psi = \pi^{1/3}(6V)^{2/3}/A$ (Wadell, 1932),

where V is the particle volume and A its surface area. Bounded above by unity (a sphere); typical ballast values lie in the range 0.68–0.80.

tracer particle

A ballast particle individually marked (here, with a small acrylic paint patch and a hand-written number) so that it can be recovered after a test and matched, one-to-one, with its pre-test 3D scan. In this thesis, five tracer particles are used per test, giving 45 tracers across the nine-cell factorial matrix.

transition zone

A region of track where the support stiffness changes abruptly, such as the approach to a bridge, a tunnel portal, or a level crossing. The stiffness mismatch concentrates impact loading and is a known accelerator of ballast degradation. Transition zones are the principal use case for Under-Ballast Mat (UBM) reinforcement.

Tukey $1.5\times$IQR fence

A robust outlier-detection criterion that flags a data point as an outlier if it falls below $Q_1 - 1.5 \cdot \mathrm{IQR}$ or above $Q_3 + 1.5 \cdot \mathrm{IQR}$, where Q_1 and Q_3 are the first and third quartiles. Formally introduced in Appendix C; used in Chapter 5 to identify catastrophic Mode B fracture events.

Notation

Symbols used throughout the thesis are listed below, grouped by theme. SI units follow the conventions of `siunitx`; dimensionless quantities are marked [–]. Acronyms (e.g. PUBM, RUBM, FIIN, RDSO, STL) are listed in the *List of Abbreviations*.

Apparatus and loading

Symbol	Meaning	Units
m	Hammer mass	kg
h	Hammer drop height (above pad)	mm or m
g	Gravitational acceleration ($= 9.81$)	m/s^2
K_e	Kinetic energy of hammer at impact, mgh	J
D	Mould internal diameter	mm
A_p	Cross-sectional area of pad/specimen	mm^2
F	Peak hammer force (in-line load cell)	kN
σ	Nominal contact stress at specimen surface, F/A_p	kPa
N	Cumulative number of impact blows	[–]

Bulk-specimen response

Symbol	Meaning	Units
s	Net cumulative axial settlement (post-first-blow)	mm
$\Delta s/\Delta N$	Incremental settlement per blow over a five-blow interval	mm/blow
$P_{0.075}$	Mass-percent passing the 0.075 mm sieve	%
P_{20}	Mass-percent passing the 20 mm sieve	%
FI_{IN}	Indian Railway Fouling Index, $P_{0.075} + P_{20}$	%

Per-particle geometry (3D mesh-derived)

Symbol	Meaning	Units
V	Particle volume	mm^3
A	Particle surface area	mm^2
V_{ch}	Convex-hull volume	mm^3
A_{ch}	Convex-hull surface area	mm^2
D_{eq}	Equivalent spherical diameter, $(6V/\pi)^{1/3}$	mm
L, I, S	Long, intermediate, and short bounding-box dimensions $(L \geq I \geq S)$	mm
a, b, c	Semi-axes of best-fit ellipsoid $(a \geq b \geq c)$	mm
Ψ	Sphericity $= \pi^{1/3}(6V)^{2/3}/A$ (Wadell, 1932)	[–]
κ_s	Sphericity ratio $= D_{eq}/L$	[–]
C	Convexity $= V/V_{ch}$	[–]
T	Surface-texture index $= A/A_{ch}$	[–]
E	Elongation $= I/L$	[–]
A_b/A_p	Flatness $= S/I$ (alternative form $(L+I)/(2S)$)	[–]
R	Roundness descriptor (scanner-specific)	[–]
d_{max}	Maximum gradation particle size	mm

Paired-change operators

Symbol	Meaning	Units
ΔX	Change in descriptor X: $X_{after} - X_{before}$	units of X
$\%\Delta X$	Percentage change in X: $\Delta X/X_{before} \times 100$	%
$\overline{\Delta X}$	Population mean of paired changes	units of X
$\widetilde{\Delta X}$	Population median of paired changes	units of X

Statistical quantities

Symbol	Meaning	Units
n	Sample size	$[-]$
$\bar{x}$	Sample arithmetic mean	units of x
$\widetilde{x}$	Sample median	units of x
Q_1, Q_3	First and third quartiles	units of x
IQR	Inter-quartile range $= Q_3 - Q_1$	units of x
W	Shapiro–Wilk statistic	$[-]$
T^+, T^-	Wilcoxon signed-rank sums	$[-]$
z	Standard-normal test statistic	$[-]$
t	Student's t-statistic	$[-]$
p	Statistical p-value	$[-]$
α	Significance threshold ($= 0.05$ unless stated)	$[-]$

CHAPTER 1

Introduction

The ballast is the silent partner of every railway: invisible when it works, undeniable when it fails.
— Indian Railways Board memorandum, 1962

In this chapter

This chapter introduces the central role of railway ballast as the load-spreading and drainage layer of conventional track, and sets out the engineering rationale for studying its degradation under impact loading. Section 1.1 describes the load-transfer function of ballast (Figure 1.2) and identifies the degradation processes that limit its service life. Section 1.2 sets out the problem and motivates the choice of a large-scale drop-weight impact apparatus. Sections 1.3 to 1.5 state the objectives, the scope, and the thesis roadmap.

1.1 Background: the central role of ballast in railway track

Railway ballast is a fundamental component of conventional track structures, consisting of coarse, angular aggregates—typically crushed stone—placed beneath and around sleepers to ensure track stability and performance. Its primary functions include distributing train loads from the sleepers to the underlying subgrade, providing lateral and longitudinal resistance to track movement, and facilitating effective drainage to prevent water accumulation and loss of strength in the track foundation. The mechanical behaviour of ballast is governed by factors such as particle size, shape, gradation, and strength, which influence its load-bearing capacity and resistance to degradation under repeated train loading. Over time, ballast experiences fouling and particle breakage, which can reduce its drainage capability and stiffness, leading to track settlement and increased maintenance requirements. Recent studies emphasise the importance of understanding ballast's geomechanical properties and dynamic response to improve track longevity and safety, especially under high-speed and heavy-haul conditions. Consequently, ongoing research explores alternative materials, stabilisation techniques, and advanced monitoring methods to enhance ballast performance and sustainability in modern railway systems.

The mechanical behaviour of the ballast layer is governed, ultimately, by the morphological state of its constituent grains. Freshly quarried angular particles possess interlocking micro-asperities that generate very high inter-particle friction; a well-compacted layer of such particles can carry train loads with lateral deformations of less than a millimetre per million-gross-tonne of traffic. The same layer, after a decade of service, contains particles that have been rounded by abrasion and reduced in size by fracture. The rounded assembly has lower friction, the fractured fines infill the voids and reduce drainage, and the layer settles more quickly under each successive load cycle. Track geometry degrades, the track engineer dispatches a tamping machine, and a maintenance cycle begins (Esveld, 2001; Indraratna et al., 2011). The maintenance cost of ballasted track is dominated by this cycle, and delaying the onset of measurable degradation by even a few years has first-order economic consequences for a national rail network.

Under ordinary cyclic traffic the process of ballast degradation is gradual and reasonably predictable. Under impact loading—the class of loading generated at rail joints, wheel flats, transition zones where the track stiffness changes abruptly, turnouts, and level crossings—the process is accelerated by impulse stresses that rise briefly to values well in excess of the contact and tensile strength of individual grains (Remennikov and Kaewunruen, 2008). Li and Davis (2005) reported accelerated settlement rates in the metre-scale vicinity of bridge abutments that were several times the rate in the adjoining line, attributed directly to the sharp change in support stiffness that produces momentary contact-stress amplification. It is precisely at these hot-spots of impact loading that ballast degradation is most costly, both in maintenance and in the small-probability tail of safety-relevant events.

For Indian Railways—the fourth-largest rail network in the world by route-kilometre and one of the most heavily loaded by axle-load × traffic—the engineering problem of ballast degradation is particularly sharp. The network operates a mix of 22.5 t axle-load freight, heavy-haul corridors with axle loads proposed to move beyond 25 t, high-density suburban passenger services, and, increasingly, semi-high-speed passenger traffic. The ballast specification governing this mix—Research Designs and Standards Organisation (2023)—prescribes angular, machine-crushed stone with a 20–65 mm size envelope. The in-service vertical contact stress at the ballast–sleeper interface is estimated to range from approximately 240 kPa under ordinary passenger loading to values approaching 500 kPa under the combined static and dynamic action of heavy freight and impact events. Any laboratory test that purports to characterise the degradation behaviour of Indian ballast must reproduce particles in this size range and stresses in this band; any test that does not, reports a number whose physical meaning is uncertain.

Figure 1.1. Indian Railways ballasted track in service showing the sleeper–ballast system.

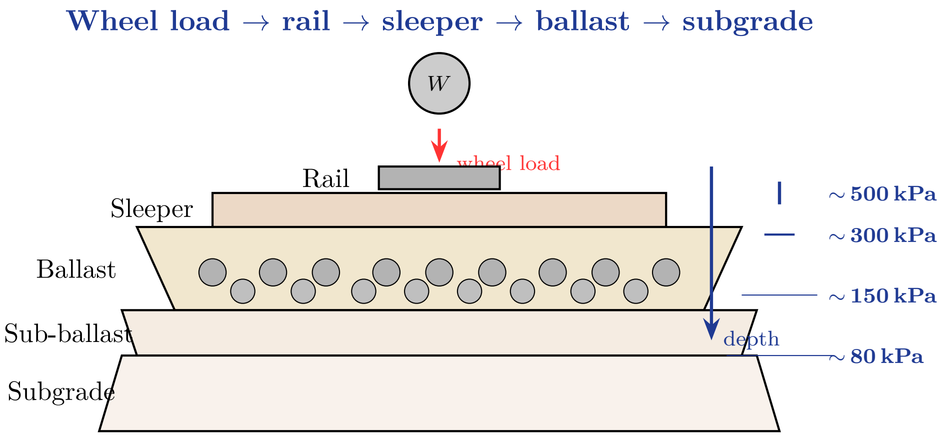


Figure 1.2. Conceptual diagram of the load-transfer mechanism in conventional ballasted track. The vertical wheel load is distributed laterally with depth, attenuating from ∼500 kPa at the top of the ballast to ∼80 kPa at the subgrade. Ballast degradation under repeated impact loading concentrates in the upper ∼150 mm of the ballast layer where the contact stress is highest.

1.2 Problem Statement

This thesis addresses two coupled gaps in Indian railway ballast assessment—gaps that reinforce each other and cannot be closed independently.

Railroad ballast, typically composed of coarse, angular aggregates, plays a critical role in providing structural support, stability, and drainage for railway tracks (Selig and

Waters, 1994). Under repeated train loading, ballast undergoes degradation through mechanisms such as particle breakage, abrasion, and fouling, leading to reduced track performance, increased settlement, and higher maintenance costs. Traditional methods for assessing ballast degradation, such as sieve analysis and manual inspection, are time-consuming, labour-intensive, and often subjective, limiting their scalability and precision. The degradation process is influenced by factors including impact loading conditions, particle morphology (size, shape, and surface texture), and environmental variables, yet these interactions are not fully understood or efficiently quantified.
The primary challenge is the lack of an efficient, objective, and automated method to assess railroad ballast degradation under realistic loading conditions. Current drop-weight impact tests simulate train-induced stresses, but standard protocols do not adequately account for variations in loading intensity, frequency, or ballast morphological changes over time (Koohmishi and Palassi, 2020). Moreover, traditional post-test analysis fails to capture the dynamic evolution of particle characteristics—such as angularity loss or fragmentation patterns—which are critical to understanding degradation mechanisms. This gap hinders the ability to predict ballast performance, optimise maintenance schedules, and design more durable track systems. There is a need for an innovative approach that integrates advanced testing conditions with analytical tools to provide a comprehensive, data-driven assessment of ballast degradation.

1.3 Objectives of the Thesis

The thesis pursues five objectives:

(i) To modify the standard drop-weight impact test apparatus (based on European standards) to replicate Indian railway ballast characteristics and loading conditions.

(ii) To evaluate the degradation behaviour of railway ballast under repeated impact loading using the modified testing setup.

(iii) To investigate the effectiveness of different reinforcement conditions in reducing ballast degradation under impact loads.

(iv) To quantify morphological changes in degraded ballast particles using 3D image processing techniques (shape, angularity, and size variations).

(v) To establish relationships between impact-induced degradation and morphological characteristics for improved assessment of ballast performance.

1.4 Scope

The scope of this thesis focuses on the experimental and analytical investigation of railway ballast degradation under controlled impact loading conditions, with particular emphasis on adapting existing testing methodologies to Indian railway requirements. The study involves modifying and calibrating a drop-weight impact test apparatus, originally based on European standards, to simulate realistic loading conditions representative of Indian Railways, including higher axle loads and varying traffic characteristics. Laboratory experiments are conducted to evaluate ballast degradation behaviour under repeated impact loading, examining parameters such as particle breakage, gradation changes, and degradation indices. In addition, the research assesses the effectiveness of selected reinforcement or stabilisation techniques in reducing ballast deterioration. A key component of the study is the application of advanced three-dimensional image-processing methods to quantify changes in ballast particle morphology, including shape, angularity, and surface characteristics before and after degradation. The thesis further aims to establish correlations between mechanical degradation and morphological evolution of ballast particles, contributing to improved understanding of ballast performance. However, the study is limited to laboratory-scale investigations, specific ballast types used in Indian Railways, and impact loading conditions, without extensive consideration of long-term field performance or environmental effects.

1.5 Thesis structure

The remainder of the thesis is organised as follows. Chapter 2 reviews the literature on ballast degradation mechanisms, laboratory characterisation under impact loading, substructure reinforcement, and three-dimensional morphological characterisation, and identifies the two gaps the present work is designed to fill. Chapter 3 documents the materials, apparatus, calibration, test protocol, and morphological-analysis pipeline. Chapter 4 reports the bulk-specimen response (settlement, peak hammer force, Indian Railway Fouling Index FI_{IN}) for the nine-configuration programme. Chapter 5 reports the paired per-particle morphological response across twenty-two descriptors for the 45 tracer particles, with a particular focus on the Mode A / Mode B dichotomy. Chapter 6 states the principal conclusions, the novel contributions, the limitations, and the recommendations for future work.

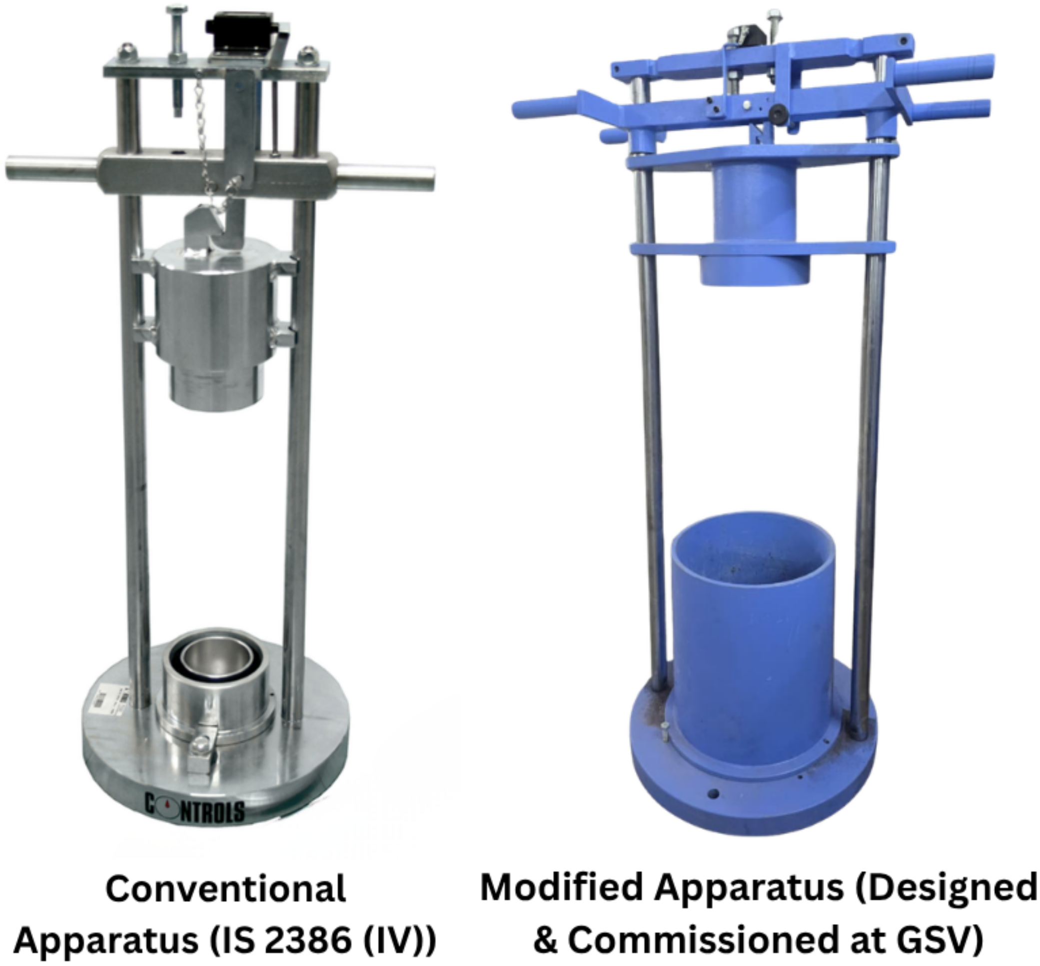


Figure 1.3. Comparison of the modified impact-testing apparatus designed and commissioned at GSV against the conventional aggregate impact value apparatus.

Chapter takeaways

- Ballast spreads wheel loads, restrains the track laterally, and drains rainwater—failure of any of these three functions triggers maintenance.
- Under repeated loading, ballast degrades both by *rearrangement* (→ settlement) and by *breakage* (→ fouling). The two processes are related but not interchangeable; both must be measured.
- Indian Railways operates with 22.5 t axle loads and dense traffic on a flexible-base substructure that is *not* the European reference case. A made-for-India impact-testing apparatus is therefore required.
- The thesis pursues two coupled lines of inquiry: (i) bulk-specimen response (settlement, peak hammer force, Indian Railway Fouling Index FI_{IN}) and (ii) per-particle morphological response (twenty-two descriptors on forty-five tracer particles, paired pre- and post-test).

- Chapter 2 now establishes what is known and identifies the two gaps the present work is designed to fill.

CHAPTER 2

Literature Review

Standing on the shoulders of others lets us see further, but only if we first agree what is shoulder and what is sleeve.

— adapted from Robert Merton

In this chapter

This chapter synthesises four decades of research on ballast degradation and uses it to motivate the present work. The literature is organised under five themes: the ballast layer in track system context (Section 2.1); degradation mechanisms (Section 2.2); laboratory characterisation under impact loading (Section 2.3); substructure reinforcement and under-ballast mats (Section 2.4); and three-dimensional particle morphology (Section 2.5). The chapter ends with a synthesis identifying the two gaps that the present thesis is designed to fill (Section 2.6).

THE literature on ballast degradation is both deep and unusually fragmented. It is deep because serious laboratory work on granular skeletons under cyclic and impact loading has been continuous since the 1960s, and it has been reinforced in the last two decades by discrete-element modelling, by field monitoring of mainline track, and—most recently—by three-dimensional imaging of individual grains. It is fragmented because these different strands have grown largely independently of one another: the cyclic-triaxial community and the impact community have different journals and different index vocabularies; the image-analysis community has only recently made sustained contact with the degradation-mechanics community; and the practitioners who set national ballast specifications have, in turn, moved at their own pace. This review is organised thematically rather than paper-by-paper in order to make sense of these strands together.

Six themes are traced in turn. Section 2.1 places the ballast layer in the track system and reviews its functional requirements. Section 2.2 discusses the mechanisms of ballast degradation that the thesis is ultimately concerned with. Section 2.3 examines the laboratory characterisation of degradation under cyclic and impact loading, with particular attention to the apparatus and the size-effect problem that motivates

Chapter 3. Section 2.4 reviews the use of resilient substructure inclusions—pads, mats, and compliant bases—to intercept and dissipate the impulsive energy that drives grain-scale damage. Section 2.5 surveys the emergence of three-dimensional morphological characterisation, which supplies the measurement framework used in Chapter 3 and applied in Chapter 5. Section 2.6 draws the preceding five sections together and identifies the two gaps—the apparatus gap and the characterisation gap—that this thesis addresses.

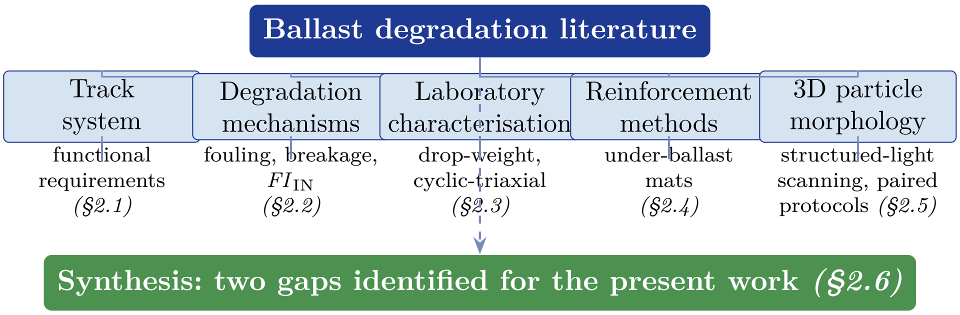


Figure 2.1. Taxonomy of the ballast-degradation literature reviewed in this chapter. The five primary themes converge in Section 2.6, which identifies two open gaps—an *apparatus gap* (no large-scale drop-weight apparatus calibrated to the Indian Railways stress envelope) and a *characterisation gap* (no paired bulk-and-grain 3D-morphology study across a factorial pad–base–energy grid)—that the present thesis is designed to fill.

2.1 The ballast layer in the railway track system

2.1.1 Functional requirements

The conventional ballasted track is a layered granular–structural system whose task is to convert the concentrated contact forces at the wheel–rail interface into diffuse, tolerable stresses at the subgrade surface. The ballast layer—a 250–350 mm deep mat of crushed, angular stone placed beneath and around the sleepers—is the principal load-distributing element of this system. Selig and Waters (1994) enumerated its six primary functions:

(i) distribution of vertical wheel loads from the sleepers to the subgrade over an area substantially larger than the sleeper footprint;

(ii) restraint against lateral and longitudinal sleeper movement through inter-particle friction and interlock;

(iii) absorption of some fraction of the dynamic energy imparted by passing trains;

(iv) provision of rapid drainage of water away from the track formation;

(v) accommodation of tamping and lining operations without structural damage;

(vi) maintenance of a resilient bearing surface that allows the rail to deflect elastically under load without plastic accumulation at the sleeper–ballast contact.

These six functions impose mutually constraining requirements on the material. High lateral restraint demands angular, machine-crushed grains with strong inter-particle interlock; high drainage demands a uniformly graded, coarse assembly with open voids; load distribution demands adequate depth and density; and long service life demands particles hard enough to resist both impact-driven fracture and cumulative abrasion. These requirements are reflected in national ballast specifications with remarkable consistency: American Railway Engineering and Maintenance-of-Way Association (2020) in the United States, UIC 719R in continental Europe, and the RDSO specification (Research Designs and Standards Organisation, 2023) in India all prescribe crushed, angular stone of comparable hardness and in a comparable 20–65 mm size band, albeit with local variations in the permissible fines fraction and in the tolerated parent-rock types.

A small but growing body of Indian-context research has begun to address the specific behaviour of RDSO-gradation ballast. Gundavaram and Hussaini (2023)—the present supervisor—demonstrated in cyclic-triaxial tests on elastomeric-polyurethane-reinforced ballast that the inclusion of a polymeric element between the ballast and the subgrade substantially reduces cumulative plastic deformation and attenuates the growth of the fines fraction. That result, on cyclic loading, motivates in part the use of the polyurethane under-ballast mat (PUBM) condition in the present impact-loading study.

2.2 Ballast Degradation

2.2.1 Fouling and its engineering consequences

The engineering consequence of the three grain-scale mechanisms is the progressive generation of fines—particles finer than 0.075 mm in the Selig and Waters (1994) usage, or finer than the RDSO 0.075 mm sieve in the Indian adaptation. These fines infill the voids of the ballast skeleton and give rise to the condition known as *fouling*. A fouled ballast has reduced drainage capacity, reduced shear strength, a softer and more plastic response to cyclic load, and a higher rate of settlement under repeated traffic (Huang et al., 2009; Tutumluer, 2013). The practical maintenance responses—ballast cleaning, partial renewal, and in extreme cases full deep-screening—are costly and disruptive, and the engineering motivation for slowing the rate of fines generation is direct and economic.

Field fouling has an exogenous contribution—coal and mineral-ore spillage from freight wagons, wind-blown dust, and subgrade pumping—that is not reproducible in a sealed laboratory mould. The laboratory characterisation of degradation therefore measures the endogenous contribution only. Within the laboratory, the self-generated fouling contribution is quantified by a family of bulk indices whose development and limitations are reviewed next.

2.2.2 Fouling Index

The Fouling Index $FoulingIndex(afterSelig\&Waters, 1994)(FI) = P_4 + P_{20}$, introduced by Selig and Waters (1994), and its metric counterpart $FI_{\text{IN}} = P_{0.075} + P_{20}$ adapted for Indian conditions by Anbazhagan et al. (2012), are weighted sums of the percentages finer than key sieves and are intended to capture the fines-generation aspect of degradation specifically rather than the total Particle Size Distribution (PSD) shift. The fouling index is dimensionless, is reported in weight percent, and is the bulk metric most directly tied to the engineering consequences that motivate ballast replacement in the field—loss of drainage, loss of shear strength, and the progressive infilling of voids that drives settlement-bowl formation at impact-prone locations. Because it isolates the fines fraction and the sub-acceptable fraction, FI_{IN} is also the most widely reported degradation metric in the Indian-ballast literature and is referenced (in its imperial form) in international ballast-fouling specifications.

$$FI_{\text{IN}} = P_{0.075} + P_{20}. \tag{2.1}$$

The structural blind spot of FI_{IN}

FI_{IN} is a scalar summary of the post-test particle-size distribution. Two test specimens can produce identical FI_{IN} values while differing in the grain-scale distribution of damage: one specimen may have experienced a small amount of abrasion distributed across all its grains, another may have experienced catastrophic fracture of a few geometrically weak grains with the remainder largely untouched. If the two damage patterns happen to push comparable fractions across the 0.075 mm and 20 mm sieves, the index reports the same value — but the two specimens have completely different underlying physical states and engineering implications.

2.2.3 Factors affecting ballast breakage and its propagation

The rate and mode of ballast breakage are not properties of the rock alone but emerge from the interaction of intrinsic material variables with extrinsic loading and boundary variables. The intrinsic variables include the parent-rock mineralogy and grain texture, the unconfined compressive and tensile strengths, the density of pre-existing micro-cracks, and—at the assembly scale—the particle size distribution and the particle shape (angularity, sphericity and flatness/elongation). The extrinsic variables include the magnitude and frequency of the applied load, the stress path (monotonic versus cyclic, confined triaxial versus impact), the moisture state, the degree of confinement provided by adjacent layers, and the stiffness of the substructure on which the ballast rests. Indraratna et al. (2011) and Sun et al. (2014) summarise these dependencies and show that, holding intrinsic properties constant, the breakage index rises sharply with peak stress and with the number of load cycles, but is moderated by confining pressure (which suppresses tensile failure at point contacts) and by the introduction of a resilient layer beneath the ballast (which lengthens the contact pulse and lowers its peak).

At the grain scale the propagation of breakage is controlled by the geometry of the contact and the distribution of pre-existing micro-flaws within the parent rock. Hertzian contact theory predicts a tensile-stress lobe just below a point or near-point contact whose magnitude scales with the contact force to the one-third power; once this tensile stress exceeds the local tensile strength, a crack initiates and propagates along the path of least resistance through the grain (Lim and McDowell, 2005). Angular grains contact each other over small areas and therefore generate higher contact stresses for a given force, which is why fresher, more angular ballast fractures preferentially in the early loading history before transitioning to an abrasion-dominated regime. Conversely, more rounded grains distribute the same contact force over a larger area, reduce the peak tensile stress, and shift the dominant degradation mode toward asperity wear. The stiffness of the substructure beneath the ballast plays an analogous role at the layer scale: a stiffer base concentrates contact forces at fewer particle–particle and particle–base contacts, while a resilient inclusion (rubber pad, under-sleeper pad, or under-ballast mat) lengthens the contact-force pulse and reduces its peak, biasing the degradation toward abrasion and away from fracture (Navaratnarajah and Indraratna, 2017; Sol-Sánchez et al., 2015).

Propagation through the assembly is governed by force-chain mechanics rather than by the average stress. Discrete-element simulations and photoelastic studies of granular media (Cundall and Strack, 1979; Tordesillas, 2007) show that load is transmitted through a sparse network of strong contacts that carry well above the mean stress, while the remaining particles bear comparatively little load. Particles that lie on a strong force chain experience contact stresses that can exceed the bulk-applied stress by an order of magnitude and are therefore the first to fracture. When such a particle splits, the chain rearranges; load is shed onto neighbouring particles, which may themselves fracture in the next loading cycle, producing a cascade until a stable contact network re-forms. This force-chain rearrangement is the mechanism by which breakage is spatially clustered rather than uniformly distributed, and it explains the field observation that fines are often concentrated in localised pockets within an otherwise comparatively intact ballast layer.

A second statistical layer concerns the variability of breakage outcomes between nominally identical particles. Single-particle crushing tests on ballast and rockfill consistently report a wide scatter in the failure load even within a narrow size band, and the distribution is well described by Weibull statistics with a modulus typically in the range 3–6 for basaltic and granitic ballast (Lim et al., 2004; McDowell, 2002). The Weibull modulus encodes the heterogeneity of internal flaws: lower moduli indicate that a small population of weak particles will fracture early in the loading history, while the bulk of the assembly continues to degrade by abrasion. This is consistent with the bimodal damage pattern observed in laboratory specimens, where a few catastrophically split particles coexist with a majority showing only corner rounding, and it sets a lower bound on how representative any single test outcome can be of the assembly response.

2.3 Laboratory characterisation of ballast under impact loading

2.3.1 Impact-loading studies and the size-of-mould problem

Impact loading—as distinct from cyclic-triaxial loading—has received less sustained attention, despite being the loading mode that dominates at impact-prone locations in service. The seminal study of Nimbalkar et al. (2012) used a 300 mm diameter drop-weight rig with a 50 kg falling hammer and a 500 mm drop height to establish that shock mats placed beneath the ballast reduce post-test bulk degradation substantially, and that the benefit of the shock mat is greatest when the underlying support is rigid. The same authors subsequently demonstrated that recycled rubber mats achieve comparable performance at much lower cost. This line of inquiry—large-diameter mould, drop-weight hammer, ballast-scale particles—is the line that the present apparatus continues.

A series of papers by Koohmishi and co-workers has extended the drop-weight methodology in several directions. Koohmishi and Palassi (2020) investigated the morphological response of basalt, granite, limestone and a mixed-lithology ballast under drop-weight impact and showed that the parent rock type governs the partition of damage between abrasion and splitting. Koohmishi and Azarhoosh (2021) and Koohmishi and Azarhoosh (2022) studied the effect of crumb-rubber incorporation on both fresh and recycled ballast, finding that 5–10 % rubber by volume reduces post-test bulk breakage measurably but that the benefit saturates—and eventually reverses—beyond about 15 %. Koohmishi and Guo (2023) combined drop-weight impact data with random-forest and support-vector-regression models and demonstrated that parent rock type and nominal impact stress together explain the majority of variance in post-test bulk degradation, with rubber content contributing an additional smaller effect. Zhang et al. (2023) and the very recent Chen et al. (2026) have continued this line on ballast–rubber mixtures, with the 2026 study being the first to explicitly couple impact-loading breakage to the gradation-evolution trajectory during a multi-blow test rather than only at the endpoint.

Across all of these studies the same apparatus-related problem appears: the conventional IS/BS aggregate impact value test (British Standards Institution, 2010; Bureau of Indian Standards, 1963) has a 102 mm diameter mould that is far too small for RDSO-gradation ballast. Marsal (1967) established that the boundary conditions of a granular test dominate the measured response until the container diameter is at least six times the maximum particle size, and Bishop (1969) independently reached a similar conclusion from work on rock-skeleton saturation response. In a 102 mm mould with 40–65 mm particles the ratio is 1.6–2.5—one is no longer testing the ballast, one is testing the mould wall. The universal response in the published impact literature has been to develop purpose-built large-diameter rigs of 250–500 mm diameter (Ferreira and Indraratna, 2023; Koohmishi and Palassi, 2020; Nimbalkar et al., 2012)—but none of these has been designed or calibrated specifically for the RDSO gradation and the Indian stress range, and none is standardised for Indian Railways quality assurance. The present thesis fills that specific apparatus gap.

Table 2.1. Representative drop-weight impact studies on railway ballast: apparatus and method comparison.

Study	Mould D	Hammer	Drop h	Variables	Metrics	Ballast
Nimbalkar et al. (2012)	300 mm	50 kg	500 mm	Shock mats, various	FI_{IN}	AU freight ballast
Koohmishi and Palassi (2020)	250 mm	40 kg	Variable	Crumb rubber, rock types	FI_{IN}	Mixed (basalt, granite, limestone)
Koohmishi and Azarhoosh (2021)	250 mm	40 kg	Variable	Crumb rubber + gradation	FI_{IN}	Recycled ballast
Koohmishi and Azarhoosh (2022)	250 mm	40 kg	Variable	Crumb rubber + recycling	FI_{IN}	Recycled ballast
Zhang et al. (2023)	∼300 mm	45 kg	450 mm	Crumb rubber, energy	FI_{IN}, image analysis	Basalt
Ferreira and Indraratna (2023)	300 mm	50 kg	450 mm	Image-based morphology	2D morphology, FI	Australian ballast
Present study	300 mm	50 kg	Variable	UR/RUBM/PUBM × FB/RB	FI_{IN} + 3D morphology (paired)	Indian ballast

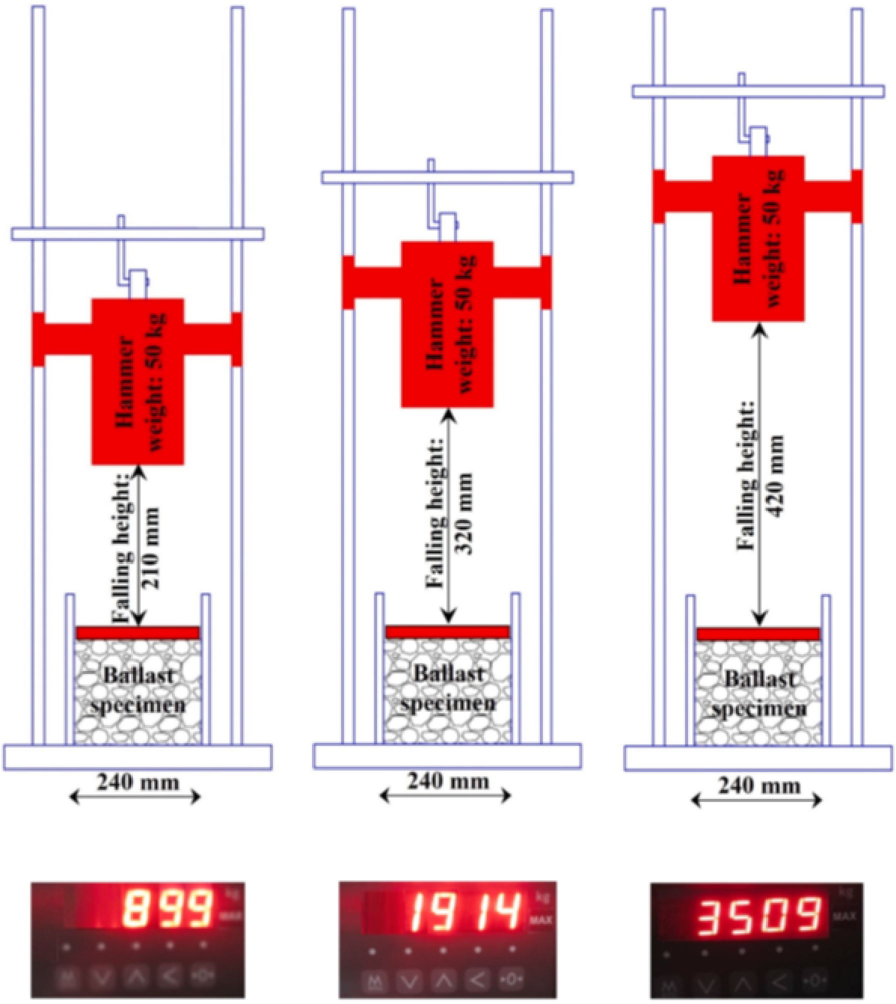


Figure 2.2. Established falling height through the impact loading test for implementing the characterised stress levels on ballast. Source: Koohmishi and Guo (2023).

Table 2.1 makes two observations explicit. First, the dominant apparatus tradition has converged on a 250–300 mm mould, a 40–50 kg hammer, and drop heights in the 300–500 mm range—all broadly consistent with the present apparatus described in Chapter 3. Second, the metrics reported are overwhelmingly sieve-based bulk metrics, with image-based morphology appearing only in the two most recent studies and in neither case as a paired before–after protocol on individually marked tracer particles. The present thesis is, to the authors' knowledge, the first to combine a large-diameter drop-weight apparatus calibrated for RDSO gradation with a fully paired per-particle three-dimensional morphological characterisation across a factorial grid of pad type, base condition and impact energy.

2.4 Ballast Reinforcement Methods

Three categories of inclusion are commonly used to reinforce or protect the ballast layer: under-sleeper pads (Under-Sleeper Pads (USPs)), which sit between the sleeper and the ballast; under-ballast mats (UBMs), which sit beneath the ballast and above the sub-ballast or subgrade; and geogrids, planar polymeric grids embedded within or beneath the ballast that confine particles through aperture interlock. The first two add compliance at a stiffness-mismatched interface to attenuate impact energy, whereas geogrids reinforce the ballast layer mechanically by restraining lateral spread of the granular skeleton.

Their mechanical regimes differ in kind. USPs are stiff elastomeric layers bonded to the underside of the sleeper that reduce sleeper–ballast contact stress, with static stiffness typically in the 0.10–0.30 N/mm^3 band (Jayasuriya et al., 2019). UBMs are placed beneath the ballast to attenuate impact energy transmitted to the subgrade; rubber variants cover roughly 0.02–0.10 N/mm^3 (Navaratnarajah and Indraratna, 2017; Ngo et al., 2019), and the polyurethane UBM (PUBM) used in the present study spans a wider product range. Geogrids, by contrast, are planar polymeric grids embedded within or beneath the ballast that confine particles through aperture interlock; they slow lateral spread of the granular skeleton and the rate of fines generation under repeated loading. Choosing among these three is therefore a question of matching mechanism to the dominant local issue—stress reduction at the sleeper interface, energy attenuation above the subgrade, or particle confinement within the ballast itself—rather than of selecting the most compliant element available.

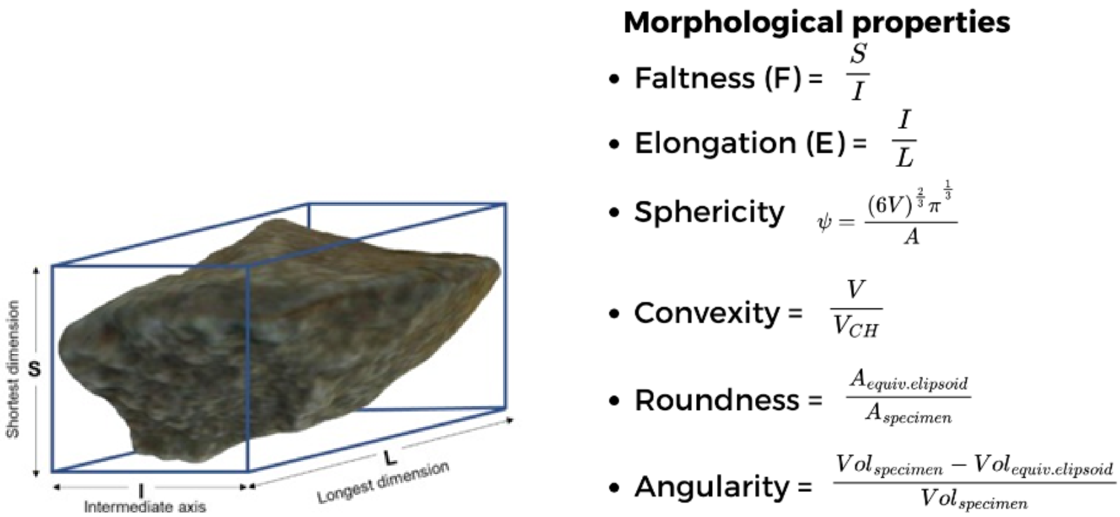
2.5 Three-dimensional morphological characterisation of ballast

2.5.1 From sieve-based to shape-based descriptors

The classical sieve-based characterisation of a ballast sample—its PSD and any index derived from it—does not resolve particle shape. Two assemblies with identical PSDs can differ substantially in the angularity, elongation, and surface texture of their grains, and the difference in shape is known to affect both initial strength and degradation kinetics. The classical shape-analysis literature begins with Wadell (1932), who introduced the concept of sphericity Ψ as the ratio of the surface area of a volume-equivalent sphere to the actual surface area of the particle:

$$\Psi = \frac{\pi^{1/3}\,(6V)^{2/3}}{A}, \tag{2.2}$$

and with Krumbein (1941), who extended the framework to sedimentary particle populations.



(a) Ballast scanning methodology. **(b)** Principal S, I, L dimensions.

Figure 2.3. Morphological parameters and the S, I, L bounding-box dimensions of a ballast particle.

2.5.2 The rise of structured-light scanning

The second-generation shape-characterisation moved from manual calliper measurement to digital imaging. Sun et al. (2014) applied three-dimensional laser scanning to the size and shape of as-supplied quarry ballast and produced one of the first systematic datasets of paired particle volumes and surface areas. Qian et al. (2014) coupled image analysis to Los Angeles abrasion testing and showed that abrasion in the Los Angeles Abrasion (LAA) mill follows a characteristic angularity-decay trajectory that can be tracked particle-by-particle. Guo et al. (2018) refined the angularity-index concept and, with the same LAA methodology, demonstrated that the size- and shape-dependence of degradation can be quantified at the single-particle level.

The tool-chain that supports this line of work has matured rapidly. Angelidakis et al. (2021) released an open-source shape-analysis code (SHAPE) that computes a standardised set of form and roundness descriptors from triangulated surface meshes. The open-source Python mesh-processing library `trimesh` (Dawson-Haggerty, 2019) has become the de-facto workhorse for reading and interrogating STL files, and the scientific-computing stack NumPy (Harris et al., 2020) and SciPy (Virtanen et al., 2020) supports the downstream statistical analysis. Kwunjai et al. (2023) combined a structured-light scanning workflow with supervised machine learning to classify deteriorated ballast and demonstrated that grain-scale shape descriptors carry the information required to distinguish fresh from service-aged ballast. Kwunjai et al. (2024) simplified the analysis pipeline while retaining the essential descriptor set.

2.5.3 Paired before–after protocols

The crucial methodological step—and the one that the present thesis is built on—is the paired before–after protocol. If the same physical grains are characterised before and after a loading event, then each grain serves as its own pre-test control and the particle-to-particle variability that is the dominant source of scatter in unpaired datasets is eliminated. The paired protocol has been used occasionally in the literature, most notably by Ferreira and Indraratna (2023) for two-dimensional image analysis of ballast under impact loading and by Bian et al. (2023) for in-situ image-aided analysis of ballast particle movement along a high-speed railway. It has not, however, been applied in combination with a large-diameter drop-weight apparatus across a factorial grid of pad type, base condition, and impact energy, which is the methodological novelty of the present study.

2.6 Synthesis and identification of gaps

Three synthesising observations emerge from the preceding five sections, and together they identify the two gaps that this thesis is designed to fill.

2.6.1 Synthesising observations

(1) The dominant laboratory apparatus tradition for impact loading of ballast has converged on 250–300 mm mould, 40–50 kg hammer, 300–500 mm drop-height rigs that satisfy the $D/d_{\max} \geq 6$ boundary criterion. However, no apparatus in the public record has been designed or calibrated specifically for the RDSO 20–65 mm gradation and the 275–525 kPa stress range representative of Indian Railways—the very specification and stress band that governs acceptance of ballast for the Indian network.

(2) The dominant characterisation tradition—the Fouling Index, in its imperial and metric forms—is bulk in character and cannot, by construction, distinguish diffuse abrasion from catastrophic single-particle fracture. Two specimens with identical FI_{IN} values can have very different grain-scale damage patterns, with different implications for settlement-bowl formation and maintenance scheduling.

(3) The emerging three-dimensional morphological tradition has demonstrated that per-particle before–after characterisation is technically feasible and methodologically sound, but has not yet been combined with a large-diameter drop-weight

apparatus across a factorial grid of pad type, base condition, and impact energy—the very grid that is most relevant to the practical design of resilient inclusions at impact-prone locations on a national rail network.

The two gaps closed by this thesis

Apparatus gap. No published drop-weight rig has been calibrated specifically for RDSO ballast and the Indian 275–525 kPa stress band. Chapter 3 addresses this by designing, fabricating, and commissioning such an apparatus at GSV.

Characterisation gap. No published study has paired the large-diameter drop-weight apparatus with a per-particle 3D morphological protocol across a factorial pad–base–energy grid. Chapter 5 closes this gap with 45 tracer particles across nine configurations.

Chapter takeaways

- Ballast degradation has been studied for sixty years, but the literature is fragmented: cyclic-triaxial, drop-weight-impact, discrete-element, and image-analysis communities have moved largely independently.
- For the in-service Indian Railways stress band ($\sim$ 275–525 kPa), no published large-scale drop-weight apparatus is calibrated to the RDSO 20–65 mm gradation.
- No published study has paired large-scale drop-weight impact loading with per-particle three-dimensional morphology across a factorial pad–base–energy grid.
- The thesis is designed to close both gaps simultaneously: Chapter 3 builds the apparatus; Chapters 4 and 5 execute the coupled bulk-and-particle programme.

CHAPTER 3

Materials and Methodology

If you cannot weigh, measure, or photograph it, you cannot yet engineer it.

— after Lord Kelvin, 1883

In this chapter

This chapter documents in reproducible detail the apparatus, the materials, and the protocol used in the experimental programme. Section 3.1 characterises the basalt ballast and verifies its conformity to the RDSO gradation envelope. Sections 3.2.2 and 3.2.3 specify the two under-ballast mat materials. Section 3.4 describes the modified large-scale drop-weight impact apparatus (Figure 3.1). Section 3.5 calibrates the apparatus against the Indian Railways field-stress envelope. Section 3.6 fixes the assembly and blow protocol; Section 3.6.3 sets out the nine-cell factorial test matrix (Figure 3.13). The chapter ends with the twenty-two-descriptor per-particle pipeline (Sections 3.7 and 3.7.2).

This chapter documents the materials, the apparatus, the test protocol, and the per-particle morphological-analysis pipeline used in the experimental programme. The chapter is the longest in the thesis because the programme combines two methodological strands—bulk-index testing through a modified drop-weight impact apparatus, and paired three-dimensional morphological characterisation of individually marked tracer particles—and reproducibility of either strand requires a fully specified procedure. The chapter is organised into eight sections following the natural workflow of the programme: materials (Section 3.1), pad and base conditions (Section 3.2), tracer-particle preparation (Section 3.3), the modified drop-weight impact apparatus (Section 3.4), stress-and-energy calibration to Indian Railways field loading (Section 3.5), the blow-by-blow test protocol and bulk-index computation (Section 3.6), and the image-based three-dimensional morphological-analysis pipeline (Section 3.7).

3.1 Basalt ballast

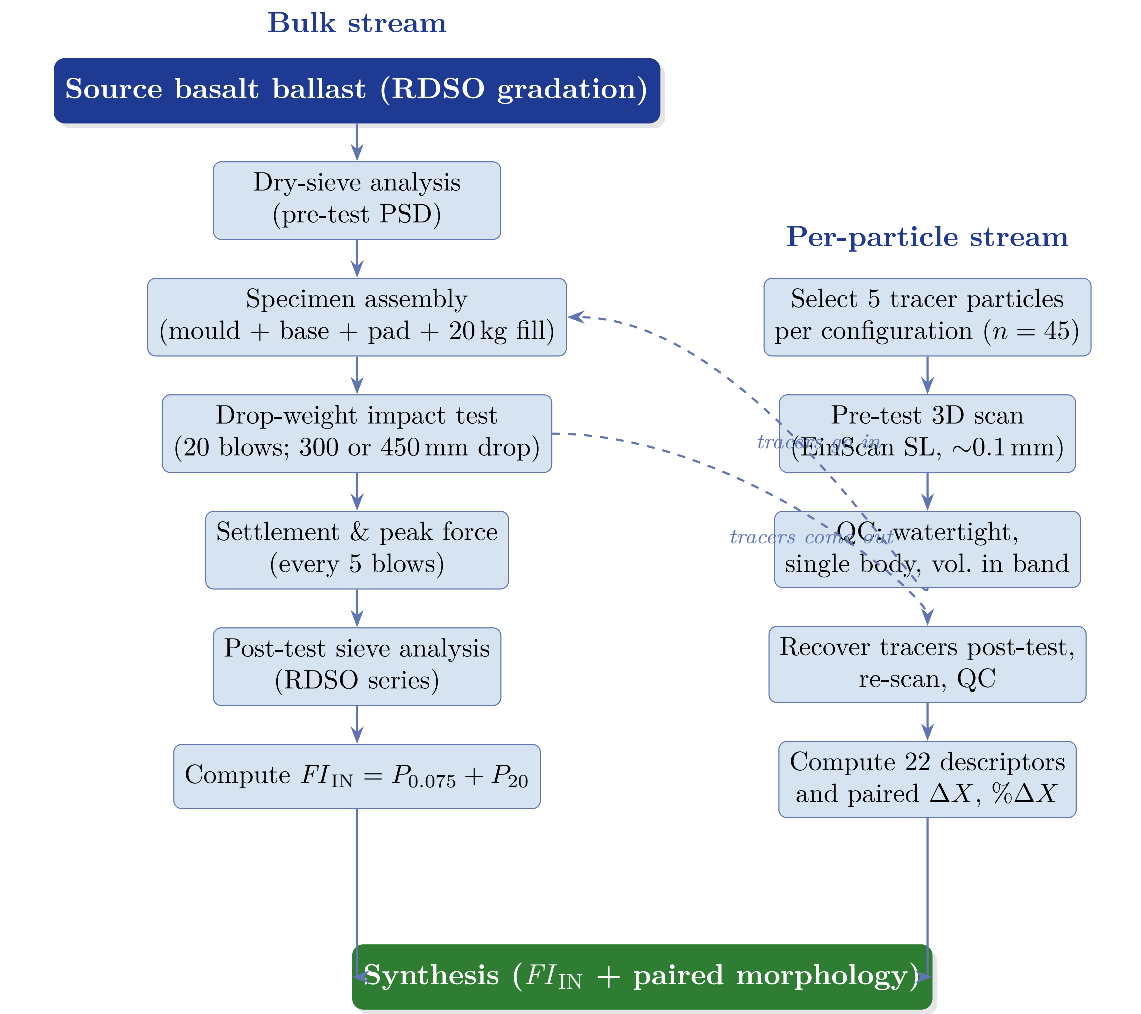


Figure 3.1. Experimental methodology: a two-stream pipeline. The bulk stream (left) follows the specimen through assembly, impact testing, and post-test sieve analysis, ending in the Indian Railway Fouling Index FI_{IN}. The per-particle stream (right) follows 45 individually-tracked tracer particles through 3D scanning, quality control, impact loading, re-scan, and 22-descriptor comparison. The two streams converge in the synthesis stage.

3.1.1 Source and parent rock

All ballast used in the experimental programme was sourced from a single quarry supplying RDSO-specification railway ballast in the Vadodara region of Gujarat, India. A single source was used to eliminate parent-rock variability as a confounding factor between tests: any cross-test difference in degradation can then be attributed to the reinforcement configuration rather than to unrecorded mineralogical variation. The parent rock is basalt—a fine-grained, dense, aphanitic igneous rock with a dry particle density 2.76 g/cm^3 and low water absorption ($< 1\,\%$). These figures are consistent with the RDSO specification (Research Designs and Standards Organisation, 2023) requirement of not less than 2.60 g/cm^3 dry density and with the parent-rock properties reported for comparable basalt in the impact literature (Koohmishi and Palassi, 2020; Zhang et al., 2023).

Approximately 25 000 kg of ballast was procured for the programme—much more than required for the nine main tests (approximately 20 kg each), for the preliminary calibration trials run during apparatus commissioning, and for a replacement reserve. All nine tests therefore drew from the same stockpile and the same parent rock. The stockpile was kept covered and elevated from direct ground contact between tests to prevent contamination of the void-fouling baseline.

3.1.2 Particle size distribution and the RDSO envelope

The as-supplied ballast was characterised by dry-sieve analysis using the following sieve series: 63, 50, 40, 31.5, 25, 20, 16, 12.5, 10, 4.75, 2.36, 0.6, 0.15 and 0.075 mm. Sieving was performed on sample masses of approximately 10 kg in accordance with the spirit of the RDSO procedure (Research Designs and Standards Organisation, 2023). The resulting gradation, plotted in Figure 3.2 against the RDSO upper and lower envelope curves, shows that the ballast conforms to the specification with a small reserve against both limits—the target gradation sits close to the mid-band of the envelope throughout the 20–63 mm range. The percentage retained on the 65 mm sieve was below the 5 % maximum permitted by the specification; the percentage passing the 20 mm sieve was below 2 %, well within the 2 % maximum fines limit. The gradation is therefore centred within the acceptable window and is representative of production-quality RDSO ballast.

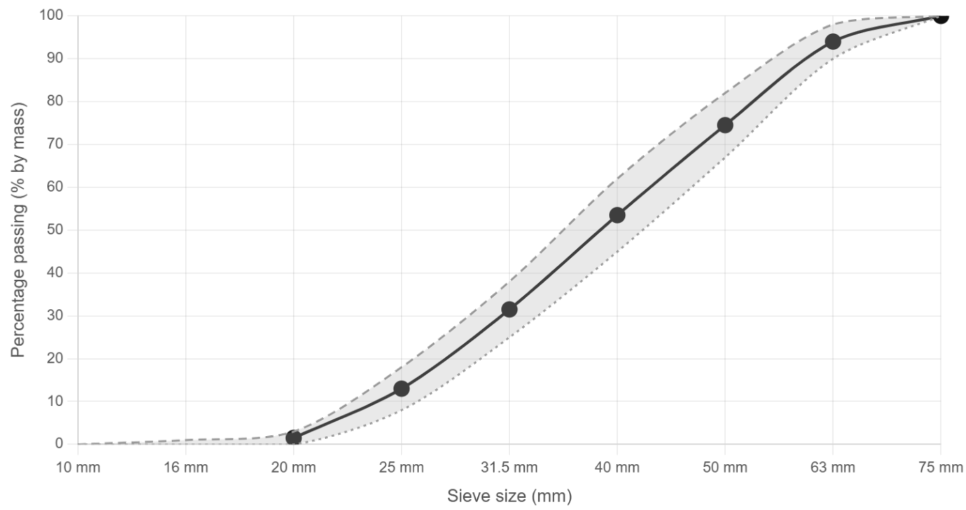


Figure 3.2. Pre-test particle size distribution of the basalt ballast used in the experimental programme, plotted against the upper and lower envelope curves of the RDSO specification.

3.1.3 Physical and mechanical properties

The ballast was characterised for the physical and mechanical properties required by the RDSO specification. The measurements are summarised in Table 3.1. All values fall within the RDSO acceptance thresholds; the Aggregate Impact Value (AIV) measured on the artificially crushed 10–12.5 mm sub-sample is reported here for completeness but is precisely the index whose limitations are set out in Chapters 1 and 2. The remainder of the programme accordingly reports the bulk fouling index FI_{IN} on the full-size specimen.

Table 3.1. Physical and mechanical properties of the basalt ballast used in the experimental programme.

Property	Measured	limit	Method
Dry particle density (g/cm^3)	2.78	≥ 2.60	Gas pycnometer + weighing
Bulk (loose) density (kg/m^3)	1460	Not specified	Bulk volume in mould
Water absorption (%)	0.72	≤ 1.0	ASTM C127 equivalent
AAV (LAA) (%)	15.4	≤ 30	IS:2386 Pt IV
AIV (%)	12.1	≤ 20	IS:2386 Pt IV (conventional)

Figure 3.3. Pile of the RDSO-gradation basalt ballast used in the experimental programme (Civil Lab, GSV).

3.2 Reinforcement Conditions

Three pad conditions and two base conditions are crossed in the factorial test matrix. The three pad conditions are an Unreinforced (UR) control, a Rubber Under-Ballast Mat (RUBM), and a Polyurethane Under-Ballast Mat (PUBM). The two base conditions are a Flexible Base (FB) of compacted sand and a Rigid Base (RB) of the concrete mould floor. The systematic three-letter abbreviations are introduced formally in this thesis as a replacement for the ad-hoc numbered test labels used in earlier laboratory work.

3.2.1 Unreinforced (UR) control

The UR configuration places no pad between the ballast fill and the base element of the mould. The UR condition serves three analytical purposes simultaneously: it is the mechanical control against which the two pad conditions are compared; it represents the in-service condition of legacy ballasted track with no substructure reinforcement; and it isolates the effect of the base condition (flexible sand versus rigid concrete) from the effect of any compliant pad placed above it. The UR condition is present in three of the nine test configurations, providing the baseline against which the pad-protected configurations are measured.

3.2.2 Rubber Under-Ballast Mat (RUBM)

The RUBM material is a commercially manufactured 12 mm thick sheet of vulcanised natural rubber with a Shore A hardness of approximately 65 and a nominal bulk density of 1.35 g/cm^3. The sheet was cut to a circular disc matching the 300 mm internal diameter of the test mould and placed flat on the base element before the ballast fill. The compliance of the RUBM falls in the middle of the range reported in the literature for natural-rubber under-ballast mats (Navaratnarajah and Indraratna, 2017; Ngo et al., 2019)—sufficient to dissipate a meaningful fraction of the hammer impulse energy while stiff enough to recover elastically between blows without progressive plastic deformation of the pad itself. No RUBM specimen was reused between tests; a fresh 12 mm disc was cut for every test to eliminate any carry-over effect of cumulative pad damage.

3.2.3 Polyurethane Under-Ballast Mat (PUBM)

The PUBM material is a commercial Getzner polyurethane under-ballast mat product. Getzner is a widely adopted supplier of polyurethane track-resilient elements on European, Australian, and Indian mainline and urban rail networks, and the specific variant used in this study is representative of the stiffness range in which Getzner UBM products are typically deployed beneath ballast in transition zones and impact-prone locations. The PUBM disc was cut to 300 mm diameter matching the mould internal diameter, like the RUBM. Gundavaram and Hussaini (2023) have previously demonstrated, in cyclic-triaxial tests on Indian ballast, that an elastomeric polyurethane inclusion materially reduces both cumulative plastic deformation and

Figure 3.4. Rubber Under-Ballast Mat (RUBM) disc, 12 mm thick natural rubber, cut to 300 mm diameter.

the rate of fines generation, and the use of PUBM in the present programme extends that line of evidence from cyclic to impact loading.

3.2.4 Base conditions (FB / RB)

The base condition—the element directly beneath the pad, or beneath the ballast if no pad is used—is the second factorial variable in the experimental design. The FB condition comprises a 100 mm deep cushion of clean, medium-grained quartz sand placed at the base of the test mould before the pad and ballast fill. The sand was oven-dried to constant mass and compacted in the mould by a standardised vibratory compaction protocol to a bulk density of approximately 1.60 g/cm^3. The RB condition places the ballast fill (with or without the interposed pad) directly on

Figure 3.5. Polyurethane Under-Ballast Mat (PUBM, commercial Getzner product), cut to 300 mm diameter.

the concrete floor of the test mould. The concrete floor is several orders of magnitude stiffer than either the pad or the sand base and is effectively unyielding at the stress levels delivered by the hammer.

The two base conditions are chosen to bracket the stiffness range that a ballast layer might encounter in service: the FB approximates a well-compacted granular sub-ballast, while the RB approximates an unyielding structural element such as a bridge deck, a concrete track-slab foundation, or a poorly-maintained subgrade that has densified below its own elastic limit. The factorial crossing of pad and base conditions allows the pad-effect contribution and the base-effect contribution to be separated cleanly, which is the analytical design that Nimbalkar et al. (2012) identified as essential for distinguishing pad performance from sub-grade compliance in laboratory shock-mat work.

3.3 Particle scanning

3.3.1 Rationale for the paired-particle design

The thesis rests on a paired-particle experimental design: the same physical grains are scanned before and after the impact-loading event, and the grain-scale damage is tracked within each particle rather than averaged across the population.

Why pair before and after?

Pairing eliminates the particle-to-particle variability in initial shape —which in any typical ballast assembly is considerable, with volumes ranging over almost an order of magnitude across the 20–65 mm gradation—as a source of scatter in the damage signal. It also enables the direct attribution of each observed change in shape descriptors to the loading event, because any change recorded in the

Figure 3.6. Test mould prepared with the 100 mm flexible base sand cushion (FB condition), photographed immediately before placement of the pad and the ballast fill.

after-scan that is not also present in the before-scan of the same particle must have arisen from the hammer blows.

3.3.2 Selection criteria

Tracer particles were selected from the bulk stockpile by visual inspection and by calliper measurement. Four criteria were applied in sequence; a candidate particle had to satisfy all four to be accepted as a tracer.

(i) **Size:** the equivalent spherical diameter—estimated at the selection stage from the three calliper-measured principal dimensions—had to fall in the 30–65 mm range.

(ii) **Shape:** particles with extreme flatness or elongation (aspect ratio $L/S > 3$ by initial calliper reading) were rejected because such shapes produce preferentially high breakage under impact loading and would distort the Mode A / Mode B distribution.

(iii) **Parent freshness:** candidates showing any visible weathering rind, iron-oxide staining, or surface cementation were rejected.

(iv) **Identifiability:** the selected particles had to admit a unique, durable surface marking for unambiguous before–after pairing.

Each selected tracer particle was individually marked with a unique code. The marking protocol uses a paint marker applied to a clean, dry surface patch on each particle. The paint coat is thin (< 0.2 mm) and does not measurably affect either the volume or the surface area computed from the scans.

3.3.3 Pre-test structured-light scanning

Each tracer particle was scanned on an EinScan (Shining 3D Technology Co. Ltd., Hangzhou, China) desktop structured-light scanner. The EinScan SE was used across the programme, providing comparable geometric accuracy of approximately 0.1 mm and scan resolution of approximately 0.15 mm in the automatic turntable mode used here. The detailed scan-protocol and mesh quality-control procedures are set out in Section 3.7 below, in the context of the morphological-analysis pipeline. In the present sub-section pre-test scanning is described as part of specimen preparation: every tracer particle carried a complete, validated, watertight triangular surface mesh in STL format into the impact test, and the pre-test mesh serves as the reference state against which the post-test mesh is paired.

3.4 The modified drop-weight impact apparatus

The apparatus comprises five principal components: a cylindrical cast-iron test mould of 300 mm internal diameter and 350 mm internal depth, seated on a rigid reaction base-plate; a vertical guide frame of four mild-steel columns connected by an upper crosshead; a cylindrical cast-iron hammer of 50 kg mass, sliding on the guide columns in free fall; a continuously adjustable hammer release-and-locking mechanism graduated over the 200–450 mm range of drop height; and a load-transfer assembly consisting of a 300 mm bearing plate, a compression-type 5 t strain-gauge load cell, and a hardened-steel striker plate. A mechanical blow counter and two manual operating handles complete the assembly. The mould diameter of 300 mm delivers a $D/d_{\max}$ ratio of approximately 6 against the RDSO 50 mm mean particle size, eliminating the boundary-dominance of the conventional IS:2386 Part IV apparatus (Bureau of Indian Standards, 1963) and bringing the present rig into the design envelope established by Nimbalkar et al. (2012) and the subsequent impact literature surveyed in Section 2.3.

The adjustable release mechanism permits the hammer to be locked at any drop height between 200 mm and 450 mm against a graduated scale on one of the four guide columns, so that the same apparatus can deliver a range of impact energies without disassembly. The in-line load cell measures the dynamic peak force imparted through the striker plate on every blow and transmits the signal to a digital peak-force indicator; this allows each test to be characterised not only by its nominal drop-energy but by the actual force time-series, which is the principal advance of the present rig over the conventional apparatus. The mechanical blow counter provides a redundant and power-independent count of delivered blows. Detailed design drawings, material specifications, tolerance callouts, and the safety-interlock documentation are set out in the utility patent application filed for this apparatus.

3.4.1 Comparison with the conventional AIV apparatus

The salient design differences between the conventional IS:2386 Part IV AIV apparatus and the modified large-scale apparatus commissioned at GSV are tabulated in Table 3.2.

3.5 Stress and energy calibration to Indian Railways field loading

The calibration of the apparatus to in-service stress conditions proceeds in two steps: first, the in-service stress range at the top of the ballast layer under Indian Railways axle loading is estimated; second, the hammer mass and drop-height range of the apparatus are chosen to span this in-service range when divided by the mould cross-sectional area.

3.5.1 Field stress estimation

The wheel load is distributed through the rail to the sleeper beneath it, and thence to the ballast over the effective sleeper–ballast bearing area. For a pre-stressed concrete mainline sleeper on Indian Railways the effective bearing area is approximately 2000–2200 cm^2, of which the two rail-seat regions each carry roughly half the axle load on each rail. Taking $A_b \approx 1100\,\mathrm{cm}^2$ as the effective bearing area below a single

Table 3.2. Principal specification of the modified apparatus in comparison to the conventional IS:2386 Part IV Aggregate Impact Value apparatus. Only the headline parameters are listed; the full specification is in the patent documentation (Appendix B).

Parameter	Conventional (IS:2386 Pt IV)	Modified apparatus (present)
Mould internal diameter	102 mm	300 mm
Mould effective depth	∼50 mm	350 mm
Hammer mass	13.5 kg	50 kg
Drop height	380 mm (fixed)	200–450 mm (adjustable)
Particle size tested	10–12.5 mm (crushed sub-sample)	20–63 mm (full RDSO gradation)
Approx. particles per test	2–4	40–80
$D/d_{\max}$ ratio	≈ 1.6 (boundary-dominated)	≈ 6 (boundary-free)
Stress simulation	Fixed / none	275–523 kPa (adjustable, maps to 22.5 t axle)
Real-time force measurement	None	5 t load cell + digital peak-force indicator
Blow counting	Manual	Integrated mechanical counter
Post-test degradation metric	AIV (% fines through 2.36 mm)	FI_{IN} + paired per-particle morphology

rail seat, the static contact stress at the top of the ballast under a 22.5 t axle is

$$\sigma_{\text{static}} = \frac{W}{A_b} \approx \frac{110.25\,\mathrm{kN}}{0.11\,\mathrm{m}^2} \approx 1000\,\mathrm{kN/m}^2 \approx 250\,\mathrm{kPa}. \tag{3.1}$$

Applying a conservative dynamic amplification of 1.5 under ordinary mainline traffic yields $\sigma \approx 375$ kPa; at impact-prone locations where the amplification can reach 2× or higher, the contact stress rises into the 500–600 kPa range. The in-service stress window at the top of the ballast layer under Indian Railways traffic is therefore conservatively taken as approximately 250–550 kPa, with the upper end reflecting heavy-haul and transition-zone conditions.

3.5.2 Apparatus stress and energy

For the modified apparatus the nominal contact stress σ is the peak hammer force F divided by the bearing-plate area A_p. The hammer kinetic energy at impact is

$$K_e = m\,g\,h, \tag{3.2}$$

where m is the hammer mass (50 kg), g is the gravitational acceleration (9.81 m/s^2), and h is the drop height. Using the measured peak forces reported at commissioning—$F \approx 19.4\,\text{kN}$ at $h = 300\,\text{mm}$ and $F \approx 37.0\,\text{kN}$ at $h = 450\,\text{mm}$—and the bearing-plate area $A_p = \pi\,(0.150\,\text{m})^2 = 0.0707\,\text{m}^2$, the two apparatus operating points correspond to

$$h = 300\,\text{mm}: \quad K_e = 50 \times 9.81 \times 0.300 = 147\,\text{J}, \quad \sigma = \frac{19.4\,\text{kN}}{0.0707\,\text{m}^2} \approx 275\,\text{kPa}; \tag{3.3}$$

$$h = 450\,\text{mm}: \quad K_e = 50 \times 9.81 \times 0.450 = 221\,\text{J}, \quad \sigma = \frac{37.0\,\text{kN}}{0.0707\,\text{m}^2} \approx 523\,\text{kPa}. \tag{3.4}$$

Calibration outcome

The two operating points—275 kPa at 147 J and 523 kPa at 221 J—correspond respectively to the lower end (ordinary mainline traffic) and the upper end (heavy-haul and transition-zone conditions) of the in-service stress window. The two drop heights selected for the test programme therefore bracket the Indian Railways field-loading envelope cleanly.

The adjustability of the drop-height mechanism over the full 200–450 mm range leaves open the possibility of finer calibration to specific axle-load classes in future work.

3.6 Sample preparation and test protocol

3.6.1 Specimen assembly

The nine test specimens were assembled using a single standardised protocol that produces reproducible specimens across the three pad conditions and the two base conditions. The protocol comprises eleven discrete steps, summarised here in the order of execution:

1. Inspection and cleaning of the cylindrical test mould.
2. Base preparation.
3. Pad placement, where applicable, with fresh discs cut for every test.
4. Tracer-particle placement at pre-assigned positions in the lower-middle of the ballast fill, with no tracer-to-tracer or tracer-to-wall direct contacts.
5. Bulk-fill placement of the remainder of the 20 kg specimen mass in two lifts.
6. Final levelling of the top surface with the load-transfer bearing-plate verification.
7. Load-cell zeroing and digital-indicator verification.
8. Photographic baseline record.
9. Hammer positioning and locking at the specified drop height.

10. Blow delivery.
11. Test termination, tracer recovery, and bulk-fill collection for sieve analysis.

3.6.2 Blow-by-blow test protocol

Each of the nine tests was conducted using an identical blow-by-blow protocol. Twenty (20) free-fall blows were delivered per test from the nominated drop height (either 300 mm or 450 mm). The choice of 20 blows is guided by two considerations. First, it is consistent with the number of blows used in comparable published drop-weight studies on ballast (Koohmishi and Palassi, 2020; Nimbalkar et al., 2012; Zhang et al., 2023). Second, preliminary trials showed that the settlement–blow curve flattens substantially between blow 15 and blow 20 and enters a near-stable zone beyond blow 15; twenty blows therefore captures both the transient settlement regime and the early approach to the steady state.

The twenty blows were delivered in rapid succession, with an interval of 5–8 s between consecutive blows—long enough for the operators to safely re-raise and re-lock the hammer, and to record the peak-force reading, but short enough that cumulative heating of the rubber or polyurethane pad is negligible. The hammer release was triggered by two-operator manual action at the upper crosshead. After each blow the hammer was secured at the crosshead before the peak-force value was read. Axial settlement was measured at five blow milestones ($N = 0, 5, 10, 15, 20$) with a depth gauge seated on the mould rim. The digital peak-force indicator was observed and recorded for each test.

3.6.3 Test matrix and nomenclature

The nine test configurations are defined by the factorial crossing of the three pad conditions (UR / RUBM / PUBM), the two base conditions (FB / RB), and the two impact energies (300 mm and 450 mm drop heights of the 50 kg hammer, corresponding to nominal contact stresses of 275 kPa and 523 kPa as derived in Section 3.5). The nine cells listed in Table 3.3 cover the factorial with the three FB–450 mm combinations omitted because the apparatus commissioning revealed that at the 450 mm drop height the sand cushion under the FB condition would progressively lose bearing capacity through cumulative compaction over twenty blows and would thereby confound the pad-effect signal with a base-consolidation signal.

Table 3.3. The nine test configurations of the experimental programme. Systematic labels are `PAD-BASE-DROP`; the legacy laboratory labels (LH, HH) are listed for cross-reference with the raw laboratory records.

Test	Systematic label	Legacy label	h (mm)	σ (kPa)	K_e (J)	Peak F (kN)
T1	PUBM–FB–300	PUBM-FB-LH	300	275	147	19.4
T2	RUBM–FB–300	RUBM-FB-LH	300	275	147	19.4
T3	UR–FB–300	UR-FB-LH	300	275	147	19.4
T4	UR–RB–300	UR-RB-LH	300	275	147	19.4
T5	RUBM–RB–450	RUBM-RB-HH	450	523	221	37.0
T6	PUBM–RB–450	PUBM-RB-HH	450	523	221	37.0
T7	UR–RB–450	UR-RB-HH	450	523	221	37.0
T8	RUBM–RB–300	RUBM-RB-LH	300	275	147	19.4
T9	PUBM–RB–300	PUBM-RB-LH	300	275	147	19.4

3.6.4 Post-test sieve analysis

At the end of each test the full bulk fill, together with any fines recovered from the mould floor after tracer extraction, was sieved through the same sieve series used for the pre-test gradation characterisation (63, 50, 40, 31.5, 25, 20, 16, 12.5, 10, 4.75, 2.36, 0.6, 0.15 and 0.075 mm).

3.6.5 Bulk degradation indices

A single fouling-based degradation index is computed on the post-test gradation—the Indian Railway Fouling Index FI_{IN}.

Indian Railway Fouling Index. Following the adaptation by Anbazhagan et al. (2012) of Selig and Waters (1994):

$$FI_{\mathrm{IN}} = P_{0.075} + P_{20}, \tag{3.5}$$

where $P_{0.075}$ and P_{20} are the weight percent passing the 0.075 mm and 20 mm sieves respectively. The first term captures fines; the second the total fraction lost from the acceptable RDSO ballast gradation.

3.7 Image-based three-dimensional morphological analysis

3.7.1 Scan protocol and mesh quality control

The seven-step scan protocol applied identically to every particle in the programme—both pre-test and post-test—comprises:

1. Cleaning (water wash, soft-brush scrub, distilled-water rinse, oven-dry at 105 °C).
2. Mounting on a low-tack dark putty pad at the centre of the automatic turntable.
3. Calibration and ambient-light verification.
4. Multi-orientation scanning to ensure complete surface coverage.
5. Mesh reconstruction.
6. Four-check quality-control gate (watertightness, single body, mesh volume within the 10 000–250 000 mm^3 range).
7. Export of the validated mesh as a binary STL file.

3.7.2 Twenty-two morphological descriptors

Twenty-two morphological descriptors are computed per mesh in three families: nine size descriptors (V, A, D_{eq}, S, I, L, V_{ch}, A_{ch}, and the three ellipsoid semi-axes a, b, c); five form descriptors (flatness $F = S/I$, elongation $E = I/L$, aspect ratio L/S, flatness index $FI = (L + I)/(2S)$, and Zingg-triangle position); and eight roundness and texture descriptors (sphericity Ψ, convexity $C = V/V_{\mathrm{ch}}$, scanner-roundness R, angularity Ang, surface-texture index $T = A/A_{\mathrm{ch}}$, sphericity ratio $\kappa_s = D_{\mathrm{eq}}/L$, and the cross-check pair ν_{ch} and A_{ratio}). The classical Wadell sphericity (Wadell, 1932) and the Krumbein roundness concept (Krumbein, 1941) remain the two principal reporting axes for grain-scale degradation.

3.7.3 Python script

The descriptors are computed by a purpose-built Python 3 pipeline based on the `trimesh` (Dawson-Haggerty, 2019), `NumPy` (Harris et al., 2020), and `SciPy` (Virtanen et al., 2020) libraries. The pipeline executes seven sequential stages on each particle's before-and-after mesh pair:

(i) Load STL.
(ii) Run the four-check QC gate.
(iii) Compute size descriptors.
(iv) Compute form descriptors.
(v) Compute roundness and texture descriptors.
(vi) Compute paired changes (ΔX, $\%\Delta X$).

(vii) Export to a multi-sheet Excel workbook with sheets for Before, After, Changes (After − Before), and Percent Changes.

The complete script is reproduced in Appendix B and has been unit-tested against synthetic reference meshes (sphere, ellipsoid, cube).

Chapter takeaways

- The basalt ballast conforms to the RDSO gradation envelope (Figure 3.2); the two under-ballast mats (PUBM, RUBM) are identified, measured, and instrumented in Sections 3.2.2 and 3.2.3.
- The modified drop-weight apparatus delivers 275 kPa / 147 J at 300 mm drop and 523 kPa / 221 J at 450 mm drop, bracketing the Indian Railways field-loading envelope (Figure 3.11).
- The nine-cell factorial test matrix (Figure 3.13) crosses three pad conditions with two base–energy combinations; the three FB–450 mm cells are excluded by design.
- Forty-five tracer particles (five per configuration) are scanned before and after impact and reduced to a twenty-two-descriptor per-particle record by the `ballast_analyzer.py` pipeline (Appendix B).
- Both methodological strands—bulk FI_{IN} and per-particle paired morphology—can now be executed and analysed (Chapters 4 and 5).

Figure 3.7. Marked tracer particles showing the white acrylic paint patch and the hand-written numeric identifier code.

Figure 3.8. EinScan structured-light scanner (Shining 3D) in operation with a marked tracer particle mounted on the automatic turntable.

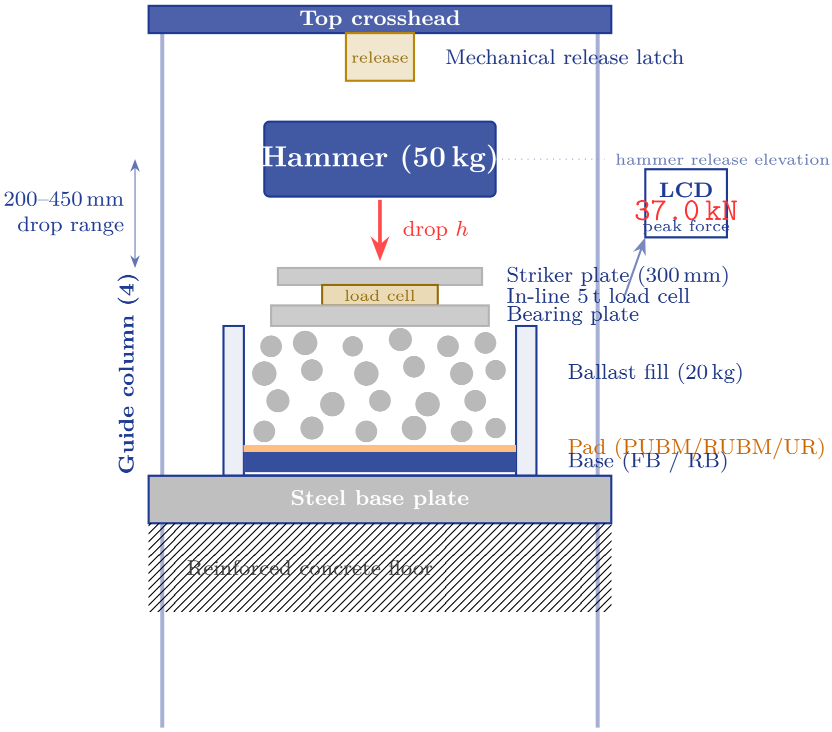
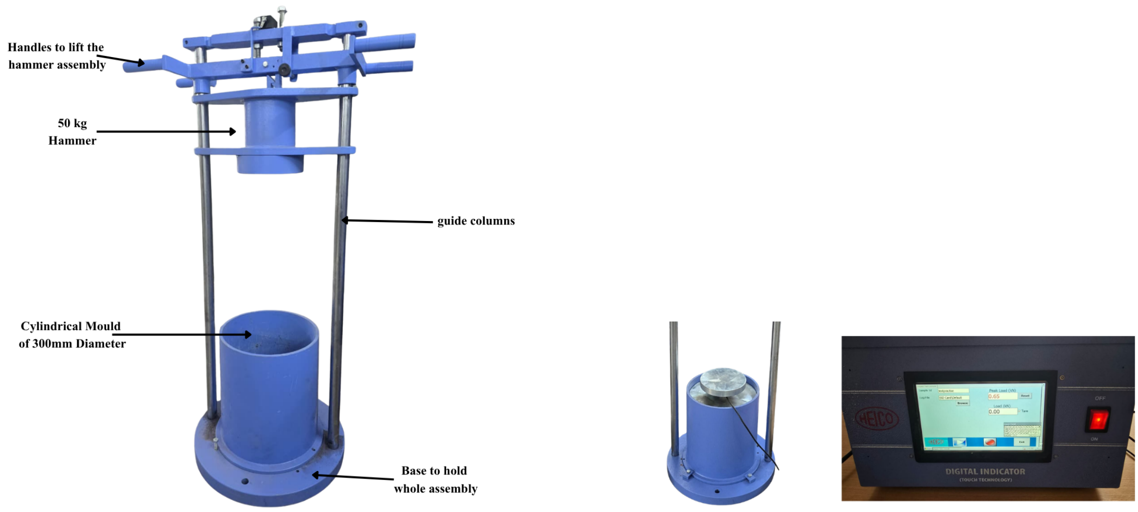


Figure 3.9. Schematic of the modified large-scale drop-weight impact apparatus. The hammer (50 kg) is locked at a release elevation between 200 mm and 450 mm above the striker plate. The in-line 5 t load cell records the peak contact force at every blow and displays the value on the LCD indicator (right). The specimen mould holds the layered base, pad, and 20 kg ballast fill. Four guide columns ensure vertical impact alignment.



(a) Full apparatus view. **(b)** Load-transfer assembly.

Figure 3.10. Modified drop-weight impact apparatus (photographs matching Figure 3.9). (a) Overall view of the apparatus and its frame. (b) Close-up of the load-transfer assembly (bearing plate, 5 t load cell, striker plate) and the digital peak-force display.

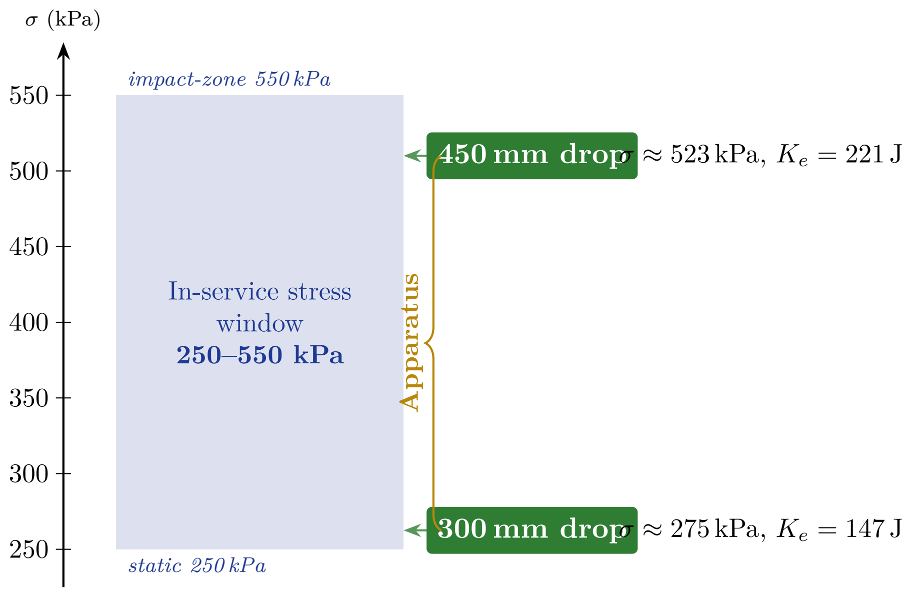


Figure 3.11. Apparatus calibration to the Indian Railways field-loading envelope. The two drop heights span the in-service stress window from ordinary mainline traffic (~275 kPa, 147 J) to heavy-haul / transition-zone conditions (~523 kPa, 221 J).

Figure 3.12. Assembling the test specimen in the mould (photograph without load-cell assembly).

	FB – 300 mm 275 kPa, 147 J	RB – 300 mm 275 kPa, 147 J	RB – 450 mm 523 kPa, 221 J
PUBM	T1 PUBM–FB–300	T9 PUBM–RB–300	T6 PUBM–RB–450
RUBM	T2 RUBM–FB–300	T8 RUBM–RB–300	T5 RUBM–RB–450
UR	T3 UR–FB–300	T4 UR–RB–300	T7 UR–RB–450

Figure 3.13. The nine-cell factorial test matrix. Rows are pad conditions (PUBM / RUBM / UR); columns are base–energy combinations. The three FB–450 mm cells are excluded by design because the sand cushion would progressively lose bearing capacity at the higher drop energy (see text).

Figure 3.14. Post-impact-test sieve analysis.

(a) Particle scanning process. **(b)** Ballast particle in EXScan software.

Figure 3.15. Three-dimensional particle scanning workflow used in the present study.

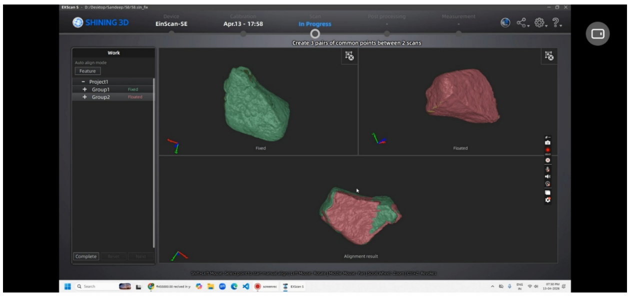


Figure 3.16. Particle aligning for multiple scans to ensure complete surface coverage.

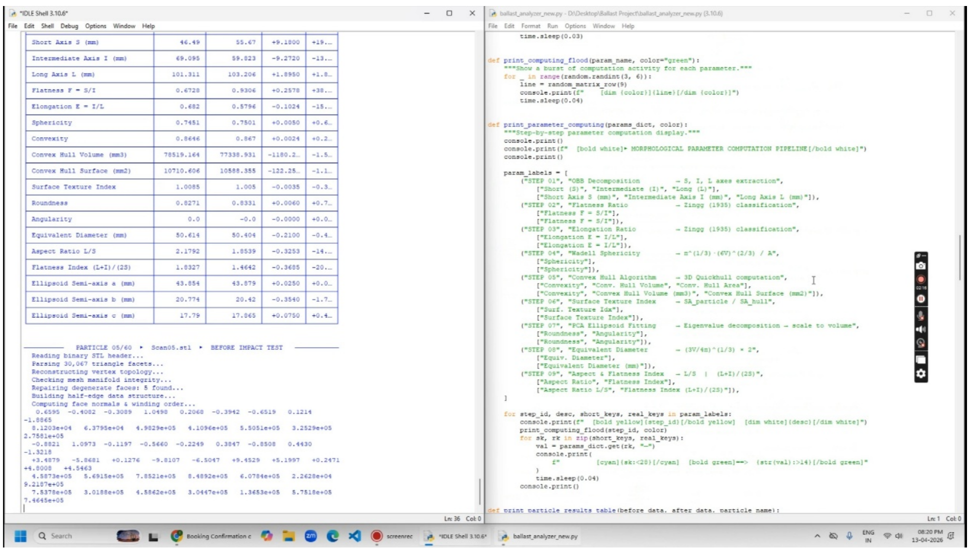


Figure 3.17. Screenshot of the running morphological-analysis script.

CHAPTER 4

Results of Impact Test Analysis

Numbers are the language in which the universe answers specific questions; the question must be specific.

— after Galileo, 1623

In this chapter

This chapter reports the bulk-specimen response of the nine-configuration programme. Section 4.1 documents the shake-down evolution of axial settlement over the twenty-blow protocol. Section 4.2 reports the peak hammer force at the two drop heights and confirms reproducible energy delivery. Section 4.3 presents the Indian Railway Fouling Index FI_{IN} for all nine tests and identifies the dominant variables. The chapter closes with a worst-case / best-case interpretation (Section 4.3.2) and a synthesis box recording the non-interchangeability of settlement and fouling (Section 4.4).

THIS chapter reports the bulk-specimen response of the nine-configuration impact-loading programme. The chapter has three purposes. First, it documents the settlement history of the four tests in which axial settlement was monitored at five blow milestones, and uses the shake-down concept to justify the choice of twenty blows as the loading endpoint. Second, it reports the peak hammer force imparted to the specimen at the two drop heights used in the programme, confirming that the apparatus delivered the nominal energy levels reproducibly across the nine tests. Third, it presents the post-test gradation analysis of the specimens through the Indian Railway Fouling Index FI_{IN} and examines the relationship between settlement and fouling.

4.1 Axial settlement

The cumulative axial settlement of the ballast specimen was recorded at five blow milestones ($N = 0, 5, 10, 15, 20$) in four of the nine tests: T1 (PUBM–FB–300), T3 (UR–FB–300), T5 (RUBM–RB–450), and T9 (PUBM–RB–300). The four settlement curves recorded are plotted in Figure 4.1, with each curve referenced to its pre-first-blow datum so that what is shown is the *net cumulative* settlement.

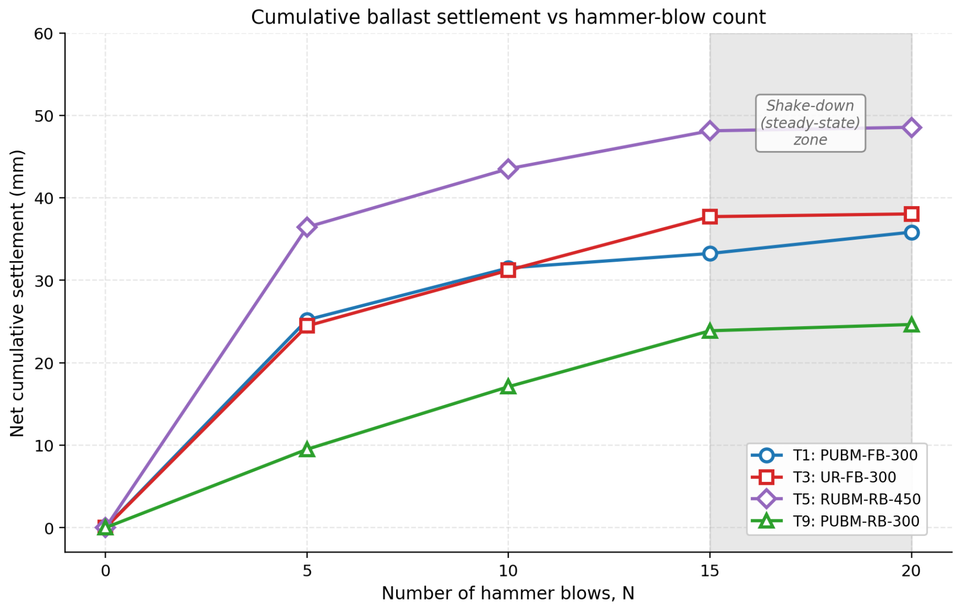


Figure 4.1. Cumulative net axial settlement of the ballast specimen as a function of hammer-blow count N for the four monitored tests. The shaded band at high N marks the shake-down (steady-state) regime within which incremental settlement per blow falls below approximately 0.5 mm.

Figure 4.2 establishes that in all four monitored tests the per-blow incremental settlement at the 15–20 interval has fallen to between approximately 0.05 mm and 0.5 mm per blow—between one and two orders of magnitude below the rate in the first interval. Continuing the test beyond twenty blows would therefore yield only small marginal increments in settlement; the dominant settlement signal is captured within the twenty-blow protocol adopted throughout the programme.

Shake-down framework

The shake-down framework used here follows the granular-material treatment of Werkmeister et al. (2001) and the impact-loading adaptation of Nimbalkar et al. (2012). Twenty blows captures the active densification regime and the early portion of the steady-state regime, which is sufficient for ranking pad and base configurations against each other.

4.2 Peak hammer force per test

The in-line 5 t load cell beneath the striker plate recorded the peak compressive force imparted on every blow of every test. Across the nine-configuration programme the peak force grouped tightly by drop height: tests run at the 300 mm drop produced peak forces around 19.4 kN per blow (corresponding to a nominal contact stress of approximately 275 kPa over the 300 mm bearing-plate area), and tests run at the

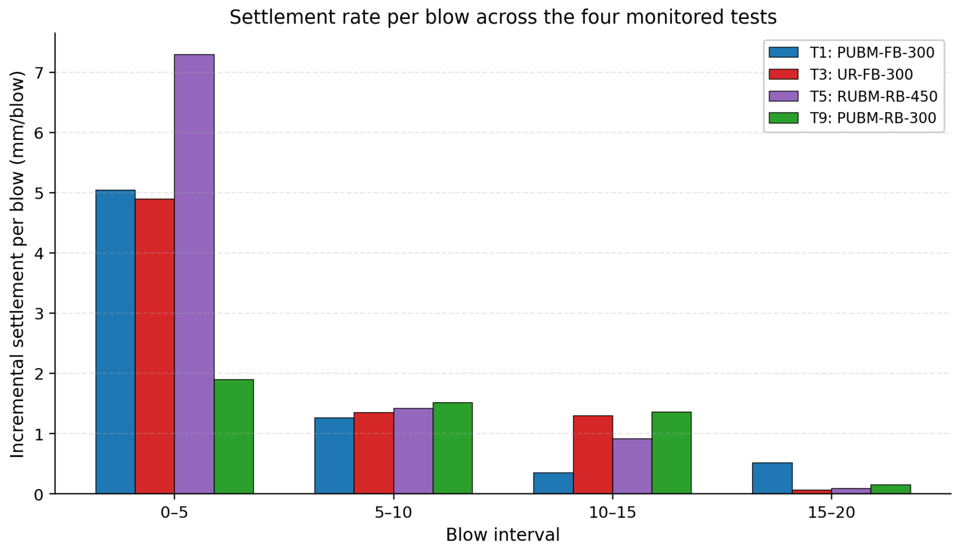


Figure 4.2. Incremental settlement per blow within each five-blow interval, for the four monitored tests. The fall in the per-blow rate from the 0–5 interval to the 15–20 interval defines the transition from active densification to the shake-down regime.

450 mm drop produced peak forces around 37.0 kN per blow (approximately 523 kPa). Figure 4.3 plots the average peak force for each of the nine tests, grouped along the horizontal axis by drop height. The two-tier structure is consistent across pad type: the presence of a UR, RUBM or PUBM pad does not appreciably change the peak force at the load cell because the pad is below the ballast-fill column and the load cell measures the through-going force at the base of the assembly, not at the ballast surface.

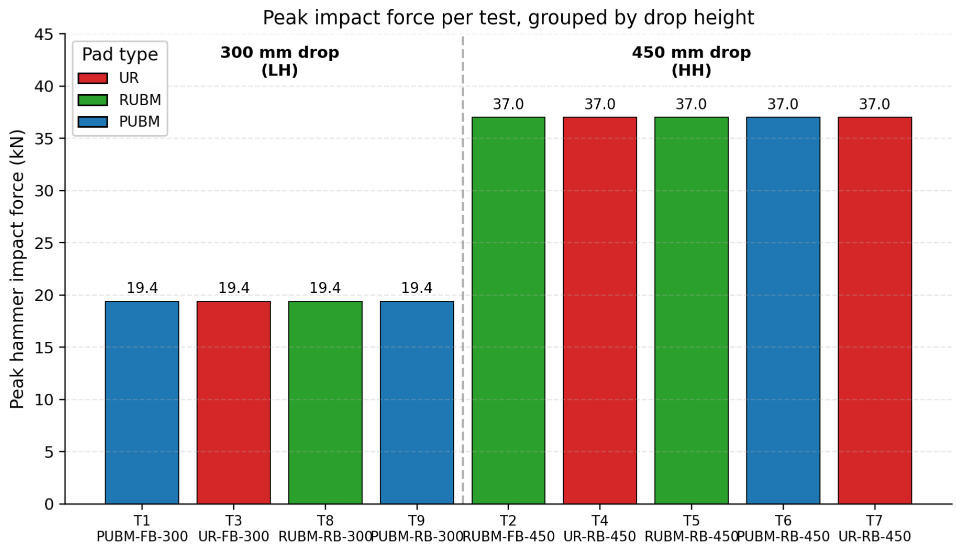


Figure 4.3. Mean peak hammer force recorded by the in-line 5 t load cell for each of the nine tests, grouped by drop height.

4.3 Indian Railway Fouling Index FI_{IN}

The post-test gradation of every specimen was determined by sieve analysis through the standard RDSO sieve series (63, 40, 31.5, 20, 16, 12.5, 10, 4.75, 2.36, 1.18, 0.6, 0.075 mm and pan). From the resulting sieve distribution the Indian Railway Fouling Index FI_{IN} was computed as defined in Section 3.6.5 and following Anbazhagan et al. (2012):

$$FI_{\mathrm{IN}} = P_{0.075} + P_{20}, \tag{4.1}$$

where $P_{0.075}$ is the percentage of post-test mass passing the 0.075 mm sieve (the fines fraction) and P_{20} is the percentage passing the 20 mm sieve (the fraction below the RDSO minimum acceptable particle size). The two terms have different physical interpretations:

- $P_{0.075}$ captures the *fines* that drive drainage degradation and the loss of inter-particle friction.
- P_{20} captures the *coarser fraction* that has migrated down out of the RDSO-acceptable envelope and that, once accumulated in the field, requires either tamping or undercutting to restore.

Both contribute to the overall ballast fouling state.

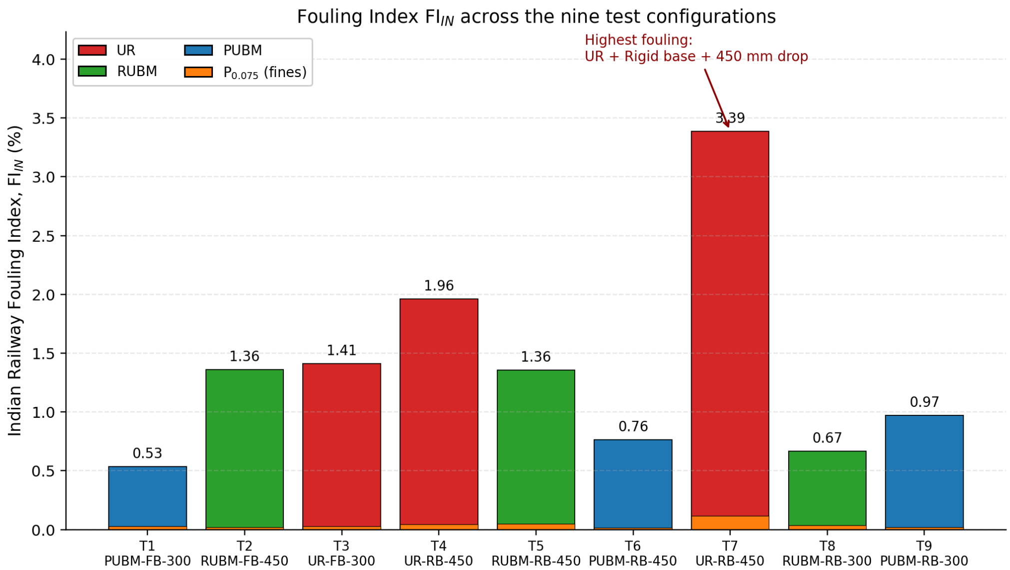


Figure 4.4. Indian Railway Fouling Index $FI_{\mathrm{IN}} = P_{0.075} + P_{20}$ (full bar) and the $P_{0.075}$ fines contribution (orange overlay) for each of the nine tests.

4.3.1 Observations

Four observations from Figure 4.4 and Table 4.1 are worth setting out explicitly.

Table 4.1. Indian Railway Fouling Index FI_{IN}, the fines fraction $P_{0.075}$, and the 20 mm-passing fraction P_{20} for the nine test configurations. Values computed per the FI_{IN} definition of Section 3.6.5 from the post-test sieve data archived in the laboratory spreadsheet. Tests are listed in systematic-label order.

Test	Configuration	$P_{0.075}$ (%)	P_{20} (%)	FI_{IN} (%)
T1	PUBM–FB–300	0.027	0.507	0.533
T2	RUBM–FB–300	0.020	1.339	1.359
T3	UR–FB–300	0.028	1.384	1.412
T4	UR–RB–300	0.046	1.914	1.960
T5	RUBM–RB–450	0.049	1.307	1.356
T6	PUBM–RB–450	0.017	0.747	0.764
T7	UR–RB–450	0.118	3.268	3.386
T8	RUBM–RB–300	0.036	0.631	0.667
T9	PUBM–RB–300	0.019	0.953	0.972

Span and dominant variables. The FI_{IN} values across the nine tests span a factor of approximately 6.4—from 0.533 % at T1 (PUBM–FB–300) to 3.386 % at T7 (UR–RB–450)—which establishes that the test-configuration variables (pad type, base condition, drop height) exert a material influence on the bulk fouling response.

Worst case: T7 (UR-RB-450). The highest-fouling configuration is T7 (UR–RB–450), the combination of no pad with a rigid base at the higher drop energy. This is mechanically the most aggressive corner of the experimental matrix—the impulsive energy is not attenuated by either a pad or a compliant base—and the result is consistent with that expectation.

Best case: T1 (PUBM-FB-300). The lowest-fouling configuration is T1 (PUBM–FB–300), the combination of a polyurethane pad with a compliant sand base at the lower drop energy. The compliant sand base absorbs a substantial fraction of the impulsive energy through frictional and re-arrangement losses, and the lower drop reduces the per-blow input energy by one third relative to the 450 mm drop.

Dominance of the P_{20} term. The fines contribution $P_{0.075}$ is small in absolute terms across the programme. It contributes between 0.017 % (T6, PUBM–RB–450) and 0.118 % (T7, UR–RB–450) of the post-test mass—only a small fraction of the total FI_{IN}, which is dominated by the P_{20} term. This indicates that the dominant bulk-level damage mechanism in the present programme is the migration of particles below the 20 mm RDSO acceptable lower bound (i.e. fragmentation that produces 16 mm and smaller daughter particles), rather than the production of fines through abrasion of asperities. The grain-scale diagnostic of Chapter 5 will revisit this distinction directly.

4.3.2 Pad-effect ranking by base condition

A further pattern that emerges from Table 4.1 is the relative ranking of the three pads on each base. The ranking is summarised in Table 4.2 and is not the same on every base–energy combination.

Table 4.2. Pad-effect ranking of FI_{IN} (%, lower is better) on each base–energy combination. The most effective pad in each row is shown in bold.

Base–Energy	Tests (PUBM, RUBM, UR)	PUBM	RUBM	UR
FB – 300 mm	T1, T2, T3	**0.533**	1.359	1.412
RB – 300 mm	T9, T8, T4	0.972	**0.667**	1.960
RB – 450 mm	T6, T5, T7	**0.764**	1.356	3.386

Table 4.2 reveals an important asymmetry. On the *rigid* base at 450 mm drop (high energy) the PUBM pad (T6, 0.764 %) outperforms both the RUBM pad (T5, 1.356 %) and the no-pad control (T7, 3.386 %); on the *flexible* base at 300 mm drop the PUBM pad (T1, 0.533 %) again outperforms both RUBM (T2, 1.359 %) and the UR control (T3, 1.412 %). However, on the *rigid* base at 300 mm drop (low energy) the ranking inverts: the RUBM pad (T8, 0.667 %) is the best performer, the PUBM (T9, 0.972 %) sits between RUBM and the UR control (T4, 1.960 %).

Pad performance is energy- and base-dependent

The polyurethane mat (PUBM) is the most effective fouling-reducer at both extreme corners of the matrix — FB at low energy and RB at high energy — but on the rigid base at low energy the softer rubber mat (RUBM) edges ahead. The implication for engineering design is that the optimal mat stiffness is itself a function of the substructure stiffness and the loading energy, and a single pad cannot be expected to be optimal across all in-service conditions. Both mats consistently outperform the unreinforced configuration on every base–energy combination.

4.4 Settlement–fouling correlation

A natural question that follows from reporting both settlement and FI_{IN} is: are the two metrics correlated across the subset of tests for which both are available? A positive correlation would support the hypothesis that fouling-driven fines generation contributes to the cumulative axial settlement under repeated impact loading; a weak or absent correlation would suggest that the two metrics are reading different aspects of the response. The correlation is therefore evaluated on the four-test settlement subset (Figure 4.5).

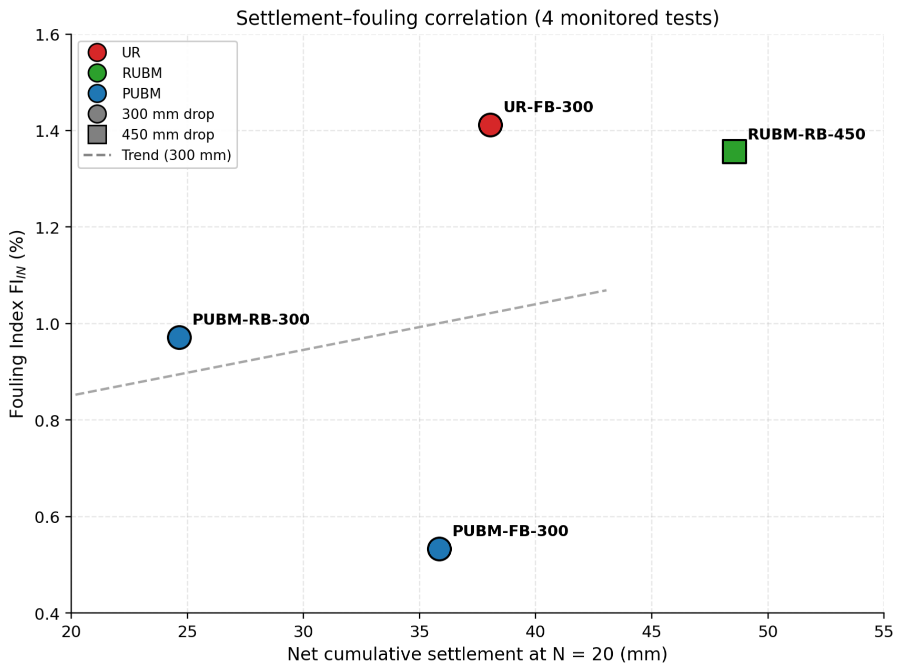


Figure 4.5. Indian Railway Fouling Index FI_{IN} plotted against the net cumulative settlement at $N = 20$ for the four monitored tests.

Two observations follow from Figure 4.5. First, within the three low-energy 300 mm tests (T1, T3, T9) a weakly-positive trend exists between settlement and fouling: the test with the largest settlement (T3, UR–FB–300 at 38.0 mm) is also the one with the largest FI_{IN} (1.412 %), and the test with the smallest settlement (T9, PUBM–RB–300 at 24.6 mm) has an intermediate FI_{IN} (0.972 %). The slope is shallow and the scatter is wide given only three points, so this trend should be read as directional rather than predictive. Second, the high-energy test T5 (RUBM–RB–450) lies clearly off the 300 mm trend: it has the largest settlement of the four (48.6 mm) but a comparatively modest FI_{IN} (1.356 %). Settlement and fouling are therefore *not* interchangeable bulk metrics; they index different aspects of the underlying degradation.

Why settlement and fouling decouple

Settlement reflects axial deformation from particle rearrangement and void closure, which need not involve fragmentation. Fouling, by contrast, requires actual particle breakage and is governed by peak contact force and parent-rock brittleness. Thus T5 settles substantially without much fragmentation, while T7 fragments without the largest settlement.

4.5 Discussion

The bulk findings of Sections 4.1 to 4.4 are broadly consistent with the published drop-weight impact literature on ballast. The shock-mat-on-rigid-base condition explored by Nimbalkar et al. (2012) produced a similar pattern of reduced bulk degradation when a compliant pad was placed beneath the ballast on a rigid support, and the natural-rubber and recycled-rubber mats in that study delivered the largest improvement on the stiffer support—directionally consistent with the present finding that PUBM substantially outperforms the unreinforced configuration on the rigid base. The crumb-rubber-modified ballast studies of Koohmishi and Azarhoosh (2021), Koohmishi and Azarhoosh (2022) and Zhang et al. (2023) reported FI fines-generation values in the same order of magnitude as the present $P_{0.075}$ results at comparable energy, and Koohmishi and Palassi (2020) reported similar fragmentation severity on basalt ballast under matched drop-weight loading. The shake-down framework used to justify the twenty-blow protocol in Section 4.1 follows the treatment of Werkmeister et al. (2001) for granular materials and is the same framework used in the impact-loading context by Nimbalkar et al. (2012).

Three patterns visible at the bulk level are worth recording before moving to Chapter 5:

(1) The PUBM-on-rigid-base configurations `T6` and `T9` broadly outperform the RUBM and UR equivalents—except at low energy on the rigid base, where RUBM (`T8`) edges ahead of PUBM (`T9`). A stiffer polyurethane mat is therefore not uniformly preferable; the optimal pad stiffness depends on the loading energy and the underlying base.

(2) The highest-FI_{IN} configuration is `T7` (UR–RB–450), the no-pad rigid-base high-energy combination—the mechanically most aggressive corner of the experimental matrix and the one most representative of the unprotected rigid-substructure conditions encountered at bridge approaches and welded-rail transition zones in service.

(3) The settlement and fouling responses are not interchangeable: the test with the largest settlement (`T5`) is not the test with the largest fouling (`T7`). Both metrics are needed to characterise the bulk response, and even together they leave open the question of which grain-scale processes drive each metric in each configuration. That is the question addressed in Chapter 5.

Chapter takeaways

- All four monitored tests reached the steady-state (shake-down) regime within the twenty-blow protocol; the chosen endpoint captures the dominant settlement.
- The peak hammer force at 300 mm drop is 19.4 kN ($\sigma \approx 275$ kPa); at 450 mm drop it is 37.0 kN ($\sigma \approx 523$ kPa). Energy delivery is reproducible across the nine tests.
- FI_{IN} spans a 6.4-fold range, from 0.533 % (T1, PUBM–FB–300) to 3.386 % (T7, UR–RB–450).
- Pad performance is energy- and base-dependent: PUBM dominates at both extremes, but RUBM edges ahead on the rigid base at low energy.
- Settlement and fouling rank the nine tests differently—each measures a different aspect of degradation. The grain-scale mechanism distinguishing them is investigated in Chapter 5.

CHAPTER 5

Results of Morphological Analysis

The grain remembers what the bulk forgets.
— laboratory adage

In this chapter

This chapter zooms from the bulk specimen down to the individual ballast grain. Section 5.1 reports the paired volume-change distribution across all 45 tracer particles and applies the Tukey 1.5×IQR criterion to identify outliers. Section 5.2 reports the paired evolution of four principal shape descriptors (sphericity, convexity, angularity, roundness). Section 5.3 synthesises the global pattern: a universal diffuse-abrasion regime (Mode A) is detected in every configuration, while a rare catastrophic-fracture regime (Mode B) is detected *only* on rigid-base configurations (Figure 5.2).

This chapter reports the results of the paired three-dimensional morphological analysis of the 45 tracer particles—five per configuration across the nine configurations of the experimental programme. The chapter is organised in four sections. Section 5.1 reports the per-particle volume-change distribution and identifies a small set of outlier particles using the Tukey inter-quartile-range criterion. Section 5.2 reports the paired evolution of four principal shape descriptors—sphericity, convexity, angularity, and roundness—before and after the impact loading. Section 5.3 examines the global pattern of mean change across all twenty-two morphological descriptors using two heat maps.

5.1 Per-particle volume change and outlier identification

Across the 45 particles the $\%\Delta V$ distribution has a first quartile of $-1.21\,\%$, a third quartile of $-0.09\,\%$, and an inter-quartile range (IQR) of $1.12\,\%$. Applying the standard Tukey $1.5 \times IQR$ fence to identify outliers yields a lower fence at $-2.89\,\%$ and an upper fence at $+1.59\,\%$. Six particles fall below the lower fence and are flagged as outliers; no particle falls above the upper fence.

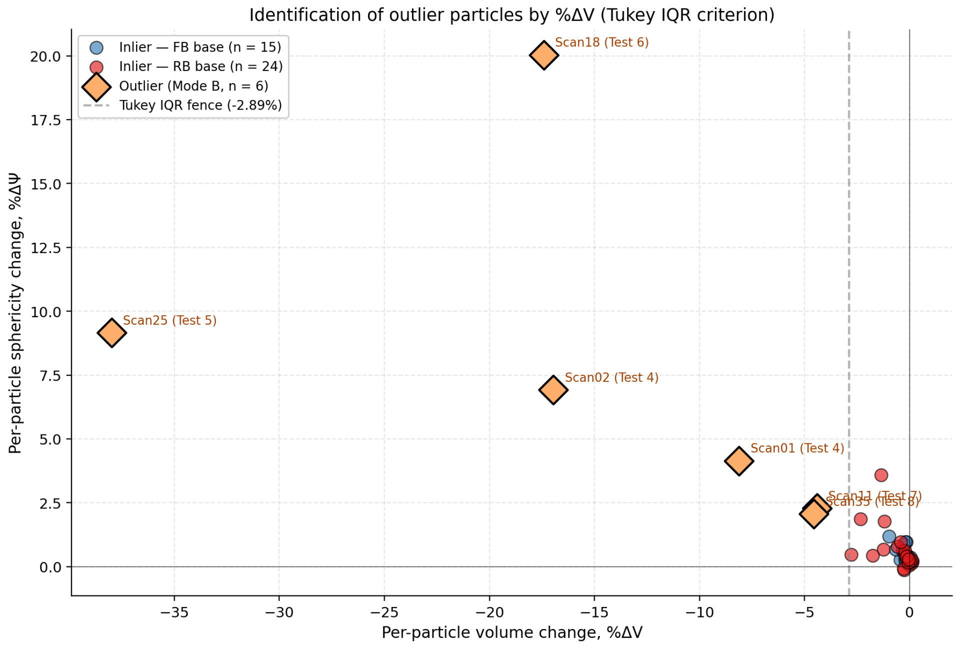


Figure 5.1. Per-particle relative volume change $\%\Delta V$ (horizontal axis) plotted against per-particle relative sphericity change $\%\Delta\Psi$ (vertical axis), for all 45 tracer particles, with outliers (Mode B) identified by the Tukey 1.5×IQR fence.

The Mode A / Mode B classification used to describe these two populations is retained as a labelling convenience throughout this chapter—Mode A refers to the 39 inlier particles (within the Tukey fence) and Mode B refers to the six outlier particles outside it—but no further mechanistic content is read into the labels themselves. The remainder of the chapter reports the morphological response of the inlier population and treats the outliers as a separately identified set whose contribution to each descriptor is quoted but not averaged into the inlier statistics.

Table 5.1. Per-test summary of the volume-change response. n_{in} = inlier count, n_{out} = outlier count, mean $\%\Delta V$ and median $\%\Delta V$ computed on the Mode A (inlier) population.

Test	Configuration	n_{in}	n_{out}	Mean $\%\Delta V$	Median $\%\Delta V$
T1	PUBM–FB–300	5	0	−0.18	−0.20
T2	RUBM–FB–300	5	0	−0.10	−0.09
T3	UR–FB–300	5	0	−0.44	−0.44
T4	UR–RB–300	3	2	−0.87	−1.24
T5	RUBM–RB–450	4	1	−0.84	−0.25
T6	PUBM–RB–450	4	1	−0.30	−0.07
T7	UR–RB–450	4	1	−0.71	−0.45
T8	RUBM–RB–300	4	1	−0.58	−0.04
T9	PUBM–RB–300	5	0	−0.18	−0.15

Where do the outliers live?

All six Mode B outlier particles are drawn from *rigid-base* configurations (T4, T5, T6, T7, T8). No outlier was observed in any flexible-base test (T1, T2, T3, T9). The compliant sand base attenuates the impulsive contact force enough to keep every tracer particle in the abrasion-only Mode A regime, whereas the rigid base allows the peak contact stress to climb past the single-particle fracture threshold in a non-negligible fraction of tracer particles.

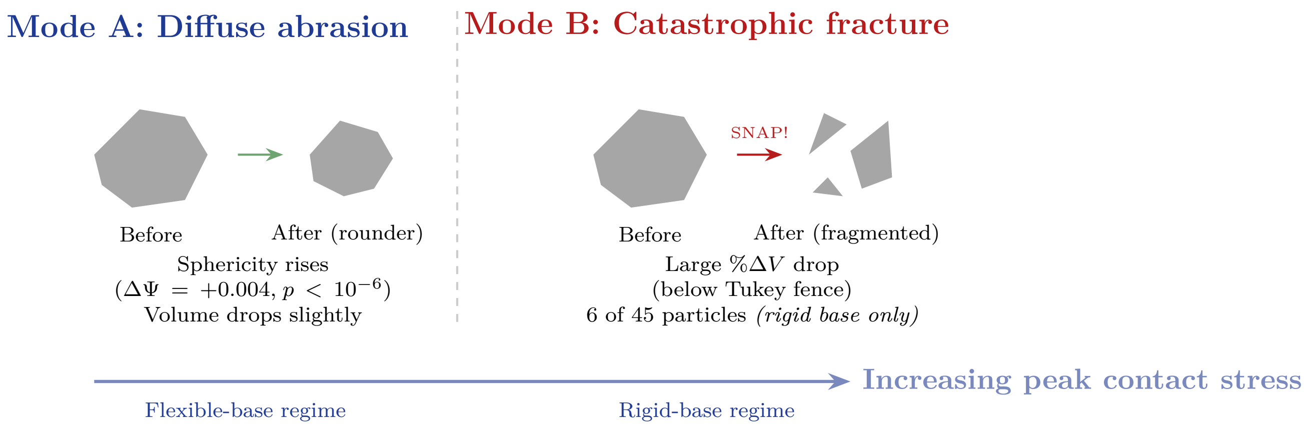


Figure 5.2. Conceptual contrast between the two damage modes identified in this thesis. Mode A (diffuse abrasion) operates in every configuration and is associated with a small but statistically significant rise in sphericity. Mode B (catastrophic single-particle fracture) was observed only in rigid-base configurations and produces large negative volume changes that fall below the Tukey 1.5×IQR fence.

5.2 Shape evolution: sphericity, convexity, angularity and roundness

5.2.1 Angularity

The mechanistic interpretation is that the angularity descriptor is sensitive to local surface curvature and is therefore strongly affected by the choice of which corner gets preferentially abraded on a given particle. A particle with two sharp corners may lose only one of them under twenty blows; the remaining corner is captured by the maximum-curvature feature in the post-test scan and the descriptor records little change, or even a small increase if the remaining corner becomes more pronounced relative to the now-smoothed neighbour. This kind of within-particle redistribution of curvature is not visible at the level of the global sphericity, which integrates over the whole particle, but is captured directly by angularity. The scatter in Figure 5.3 therefore does not contradict the smoothing-by-abrasion picture established by

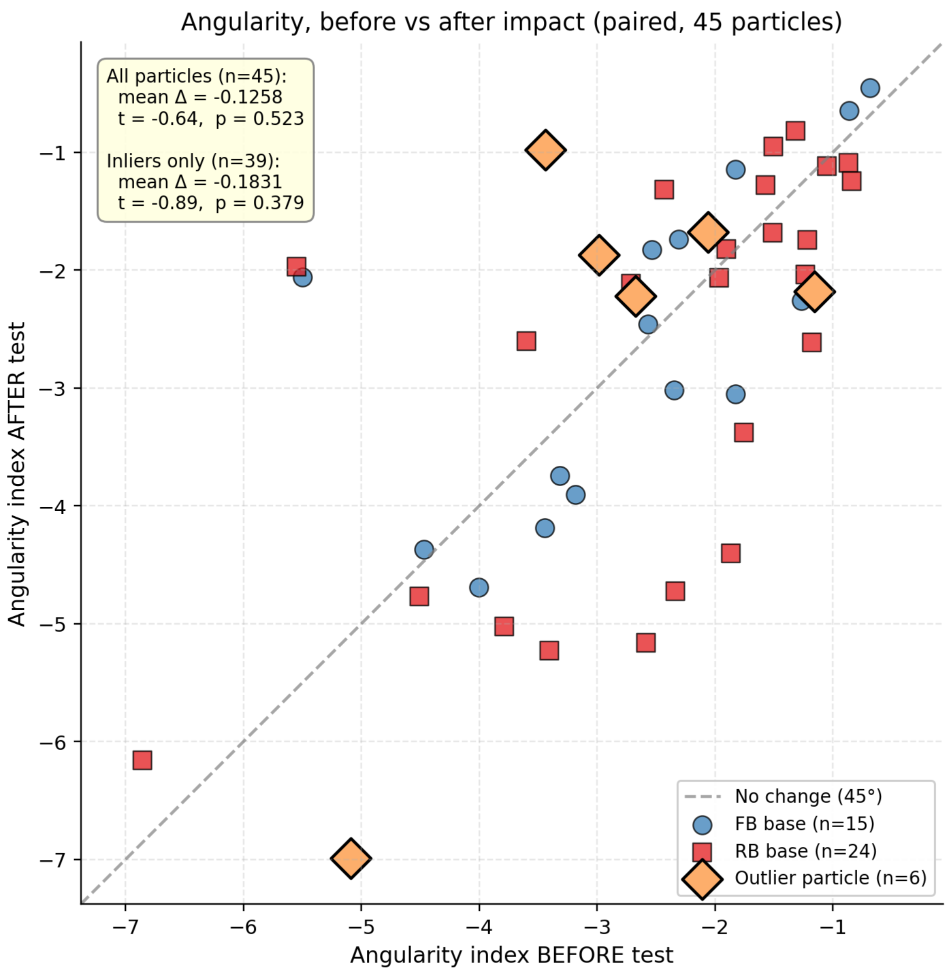


Figure 5.3. Angularity index before vs. after the impact-loading test for all 45 tracer particles. The angularity values reported here are negative scalars; less-negative (closer to zero) values indicate smoother corners.

sphericity; it adds the observation that the smoothing is locally selective, not uniformly distributed across all corners of all particles.

5.2.2 Sphericity

The sphericity—the classical Wadell descriptor (Wadell, 1932)—is the single most informative shape descriptor for diagnosing corner abrasion. A particle that loses its sharp corners without otherwise changing shape becomes more spherical. Figure 5.4 plots the paired sphericity—Ψ_{after} on the vertical axis against Ψ_{before} on the horizontal axis—for all 45 tracer particles. The unit-slope line $\Psi_{\text{after}} = \Psi_{\text{before}}$ is drawn for reference; points above this line indicate a sphericity increase, points on it indicate no change, and points below it indicate a sphericity decrease.

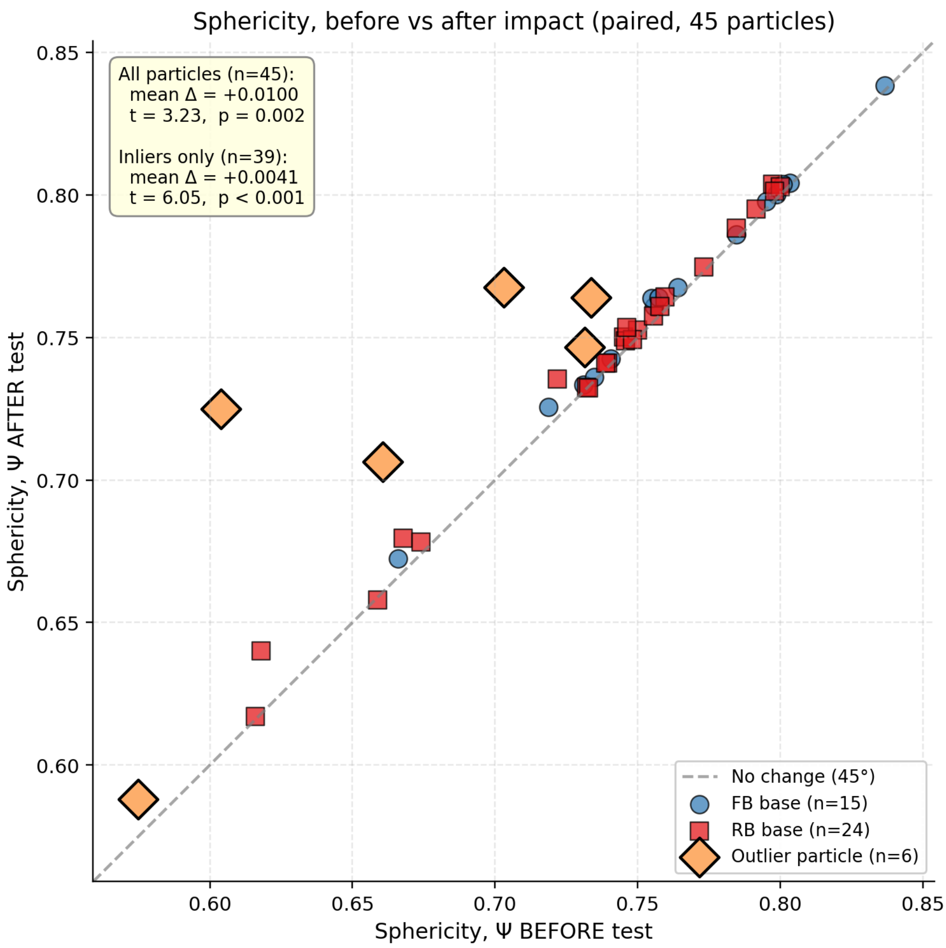


Figure 5.4. Sphericity before vs. after the impact-loading test, for all 45 tracer particles.

Universal sphericity rise

The mean paired sphericity change across all 45 particles is $\Delta\Psi = +0.0100$, and the paired t-test against the null of zero mean change yields $t = 3.23$, $p = 0.002$. Restricting attention to the 39 inlier particles only, the mean change is $\Delta\Psi = +0.0041$ with $t = 6.05$ and $p < 10^{-6}$. Sphericity rises in essentially every particle regardless of configuration — corner abrasion is therefore a *universal response* to the drop-weight impact protocol, and no configuration in the matrix (including the best-performing T1, PUBM–FB–300) eliminates it.

This is consistent with the abrasion-driven rounding observed in cyclic loading by Qian et al. (2014) and Guo et al. (2018) and in drop-weight impact loading by Koohmishi and Palassi (2020).

5.2.3 Convexity

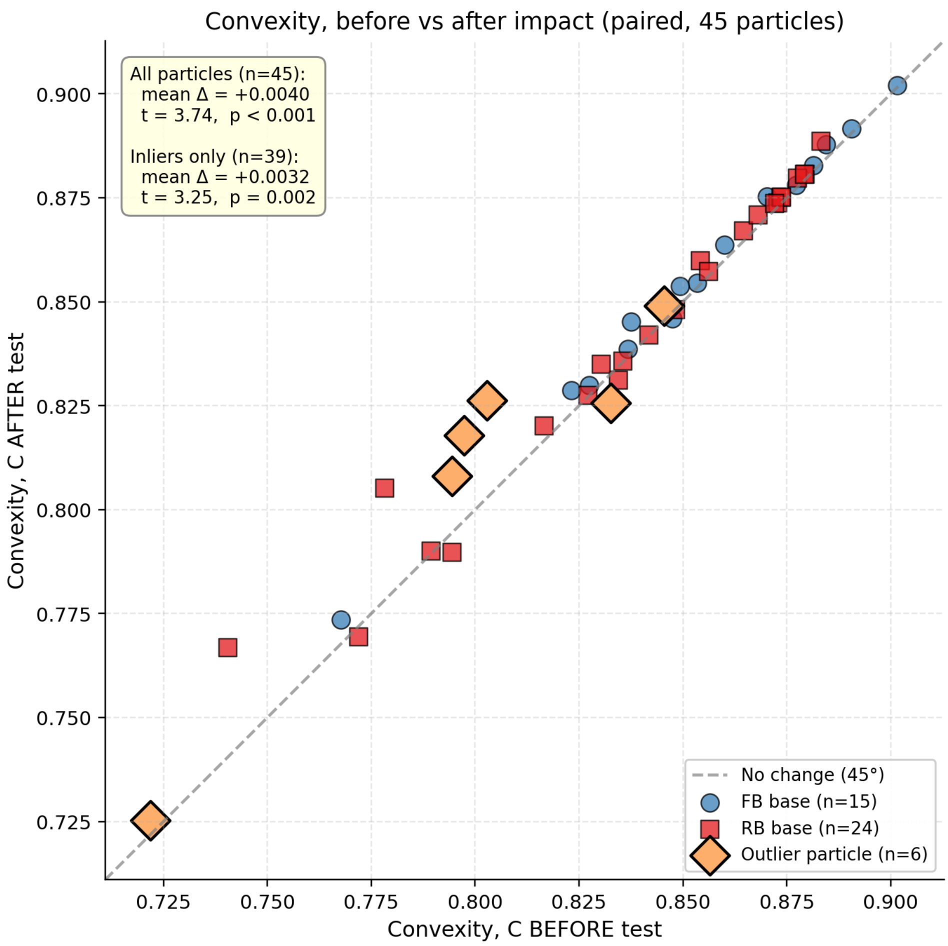


Figure 5.5. Convexity $C = V/V_{\mathrm{ch}}$ before vs. after the impact-loading test, for all 45 tracer particles. Layout and statistics as for Figure 5.4.

5.2.4 Roundness

Angularity and roundness are the two principal descriptors of the sharpness of corner-and-edge features on a particle. Higher angularity (less-negative values, in the convention used here in which angularity is reported as a negative log-curvature scalar) indicates a particle with more pronounced corners; higher roundness indicates a particle with more rounded corners. Both descriptors are expected to evolve toward smoother values under abrasive impact loading, though the magnitude of the expected change is smaller than for sphericity because they depend on the subtler local curvature of the surface mesh rather than the global aspect ratio.

The roundness response is mixed. The mean paired change is +0.038 across all 45 particles ($t = 0.67$, $p = 0.51$), and +0.058 across the 39 inliers ($t = 1.00$, $p = 0.32$). Like angularity, roundness is sensitive to local corner curvature and reflects the selective nature of corner abrasion. Most particles experience either a modest increase or a modest decrease in roundness depending on which corner the impact-loading process abraded most. The sphericity-and-convexity rise reported above is the more

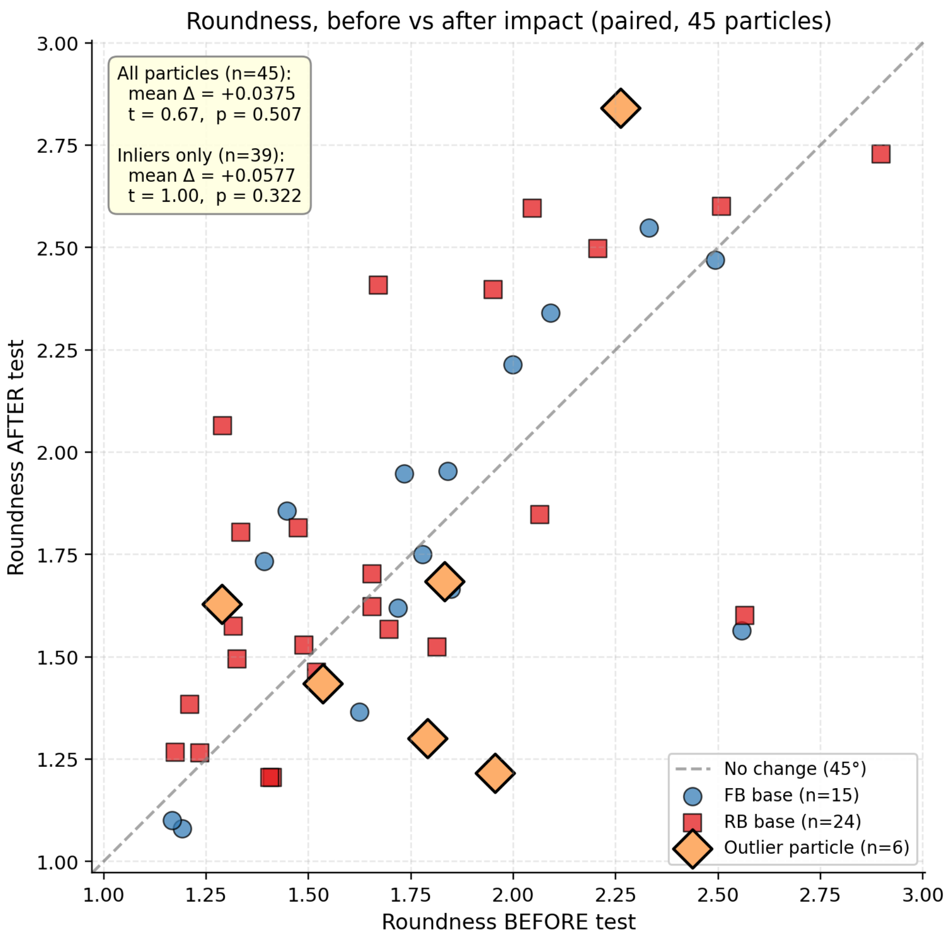


Figure 5.6. Roundness before vs. after the impact-loading test, for all 45 tracer particles.

reliable population-level diagnostic of the abrasion process; the angularity-and-roundness scatter reports the corresponding particle-level heterogeneity.

Which descriptor for which question?

For *population-level* damage diagnosis the integrated descriptors (sphericity, convexity, volume) are the robust choice. For *per-particle* damage diagnosis the locally-sensitive descriptors (angularity, roundness, surface-texture index) provide more detail but with much larger scatter and a sign that depends on which corner is abraded. Both families are needed to characterise the response fully.

5.3 Discussion: heat-map synthesis

The morphological script computes twenty-two descriptors per mesh, of which the four reported in Section 5.2 are the principal shape descriptors. To provide a global view of the pattern of mean change across all configurations and a wider set of descriptors,

Figure 5.7 collects the mean per-particle %-change of fourteen descriptors across the nine tests in two side-by-side heat maps. Outlier particles are removed before the means are computed; the visualisation is therefore a Mode A-only (inlier) picture of the morphological response. Panel (a) shows the size and primary shape descriptors, which have a comparatively narrow dynamic range (less than ±2.5 %); panel (b) shows the roundness and shape-ratio descriptors, which span a wider range (up to about ±80 %).

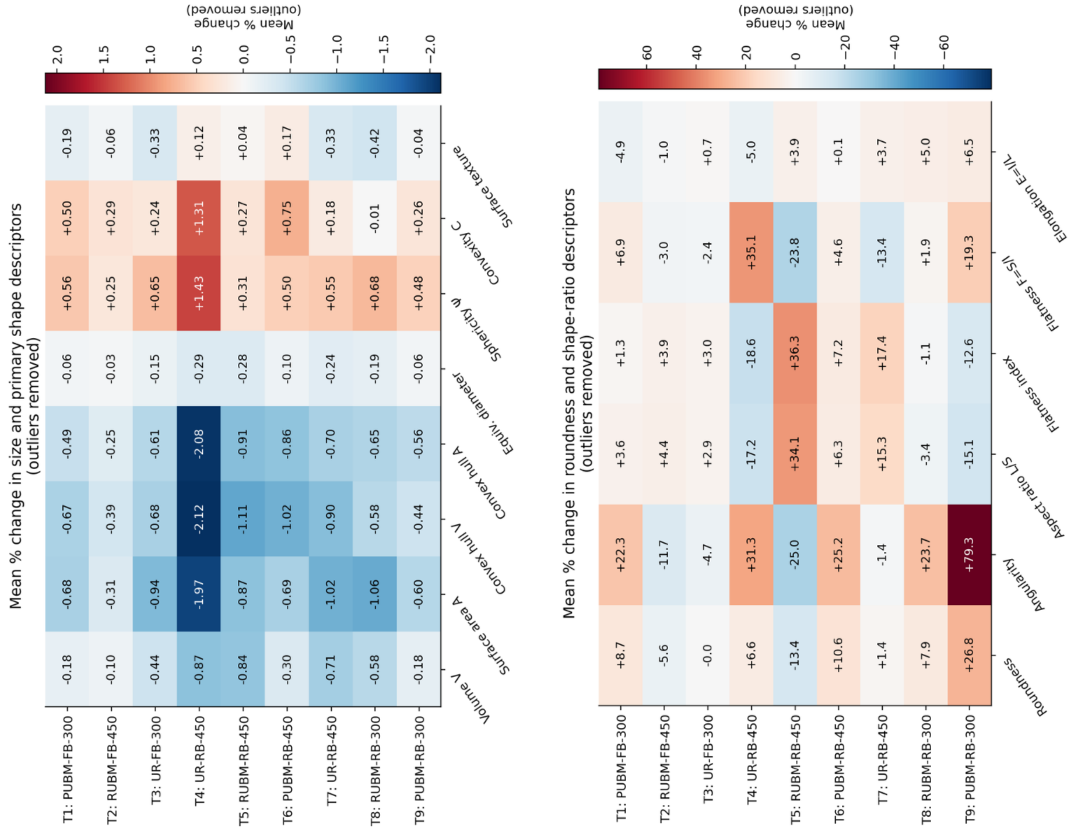


Figure 5.7. Heat maps of mean per-particle %-change in fourteen morphological descriptors across the nine tests, with outlier particles removed. Panel (a): size and primary shape descriptors. Panel (b): roundness and shape-ratio descriptors.

Three patterns are visible in panel (a) of Figure 5.7:

(1) All the size descriptors (volume, surface area, convex-hull volume, convex-hull surface, equivalent diameter) are uniformly *negative* across the nine configurations. The sign-uniformity is mechanistically reassuring: inlier particles do not gain mass under impact loading. The magnitude of the size loss tracks the ranking of Table 5.1, with the rigid-base 450 mm tests T4, T5, and T7 showing the largest size losses.

(2) Sphericity and convexity are uniformly *positive* across the nine configurations, consistent with the universal abrasion-driven rounding established in Section 5.2.

(3) The surface-texture-index column shows small but mixed-sign changes (between $-0.42\,\%$ and $+0.17\,\%$), with no clear configuration-level pattern; this descriptor is sensitive to the highest-frequency mesh asperities and is dominated by scan-to-scan reconstruction noise rather than the impact-loading signal.

Panel (b) of Figure 5.7 reports the roundness and shape-ratio descriptors, which exhibit a much larger dynamic range. The angularity column in particular spans from $-25\,\%$ at `T5` to $+79\,\%$ at `T9`—a span of more than 100 percentage points across the nine configurations. The sign of the angularity change does not group obviously by base condition or by drop height: `T1`, `T4`, `T6`, `T8`, and `T9` are all positive (sharper after the test, in the convention adopted here); `T2`, `T3`, `T5`, and `T7` are negative (smoother). The roundness column shows the same mixed-sign pattern, with the largest positive value occurring at `T9` ($+27\,\%$) and the largest negative value at `T5` ($-13\,\%$). The aspect-ratio and flatness columns are similarly heterogeneous.

Choose the descriptor to match the question

The mixed-sign pattern of locally-sensitive descriptors confirms the observation made in Section 5.2 about angularity: corner abrasion is selectively rather than uniformly distributed across the corners of each particle, and the population-level mean of these locally-sensitive descriptors is therefore not a reliable diagnostic of the configuration-level damage. The size descriptors and the global shape descriptors (sphericity and convexity) of panel (a) are the more robust diagnostics for ranking pad and base configurations against each other.

Chapter takeaways

- Mode A (diffuse abrasion) is universal: every one of the 45 tracer particles exhibits a small but statistically significant sphericity rise ($\Delta\Psi = +0.0041$, $p < 10^{-6}$, Wilcoxon).
- Mode B (catastrophic single-particle fracture) is rare (6 of 45 particles) and—critically—*occurs only on rigid-base configurations*. No outlier was observed on any flexible-base test.
- The Mode-A / Mode-B dichotomy explains the settlement-versus-fouling decoupling seen in Chapter 4: fouling is driven by the rare Mode-B events; settlement is driven by the universal Mode-A rearrangement.
- Size and global-shape descriptors (volume, sphericity, convexity) are robust diagnostics across configurations; locally-sensitive descriptors (angularity, roundness) are informative *within* a configuration but noisy *across* configurations.

- Together with Chapter 4, the chapter establishes a coupled bulk-and-grain characterisation of impact-induced ballast degradation under Indian Railways conditions.

CHAPTER 6

Conclusions and Recommendations

A thesis ends where a research programme begins.

— tradition of the doctoral oral examination

In this chapter

This chapter restates the principal contributions of the thesis, summarises the conclusions that the bulk and per-particle evidence jointly support, sets out the limitations of the present work, and recommends three immediate avenues for follow-up: an instrumented cyclic-loading extension; an in-track field-validation campaign on a heavy-haul stretch; and a discrete-element model calibrated against the paired-particle dataset reported in Chapter 5.

THIS thesis set out to address a combined apparatus-and-characterisation gap in the current assessment of railway ballast degradation under drop-weight impact loading, specifically in the context of the RDSO 20–65 mm gradation envelope and the 22.5 t Indian Railways axle-load stress range. The programme comprised: the design, fabrication, and commissioning of a purpose-built large-scale drop-weight impact apparatus at Gati Shakti Vishwavidyalaya; the execution of a nine-configuration factorial test matrix crossing three pad conditions (UR / RUBM / PUBM) with two base conditions (FB / RB) and two impact energies (300 mm / 450 mm drop); the characterisation of bulk degradation through the Indian Railway Fouling Index FI_{IN} on the post-test gradations; the paired three-dimensional morphological characterisation of 45 individually-tracked tracer particles through twenty-two descriptors before and after loading; and the statistical analysis of both characterisations individually and in synthesis. The present chapter states the principal conclusions of the programme, enumerates the specific novel contributions, identifies the limitations that bound the generalisability of the findings, and recommends concrete directions for follow-up research.

6.1 Conclusions

Six principal conclusions are drawn from the experimental programme and its analysis. They are stated here in the order in which they emerged during the research.

Conclusion 1 – Apparatus

A large-scale drop-weight setup was successfully developed at GSV, delivering consistent and reproducible impact energies (147 J and 221 J, corresponding to nominal contact stresses of 275 kPa and 523 kPa) representative of Indian Railways conditions.

Conclusion 2 – Settlement

Ballast shows rapid initial settlement that stabilises after about 15 blows, confirming a 20-blow test as appropriate. Higher energy and rigid bases cause greater settlement.

Conclusion 3 – Fouling

Fouling varies significantly with test conditions — highest under rigid base, no pad, high energy (T7: $FI_{\mathrm{IN}} = 3.386\,\%$); lowest with polyurethane pad, flexible base, low energy (T1: $FI_{\mathrm{IN}} = 0.533\,\%$). A 6.4-fold span across the nine configurations establishes that pad, base, and energy variables each exert a material effect.

Conclusion 4 – Settlement vs. fouling

Settlement and fouling measure different mechanisms — rearrangement vs. breakage — and are not interchangeable. T5 has the largest settlement (48.6 mm) but only intermediate fouling; T7 has the highest fouling but not the largest settlement. Both metrics are required for full bulk-level assessment.

Conclusion 5 – Two damage modes

Mode A (abrasion): all 45 tracer particles undergo gradual corner abrasion, increasing sphericity ($\Delta\Psi = +0.0041$ for inliers, $p < 10^{-6}$) and reducing size across all configurations.

Mode B (fracture): catastrophic particle breakage occurs in 6/45 tracer particles, and *all six are drawn from rigid-base configurations*. The compliant sand base attenuates the impulsive force enough to keep every tracer in the abrasion-only regime.

Conclusion 6 – Complementary characterisation

FI_{IN} captures bulk fouling effects, while paired per-particle morphology reveals damage mechanisms. Both are essential for a complete characterisation: FI_{IN} alone cannot distinguish diffuse abrasion from catastrophic fracture, and the per-particle protocol alone cannot speak to assembly-level fines generation.

6.2 Novel contributions

6.2.1 Apparatus and method

(1) A modified drop-weight impact apparatus for Indian ballast. A utility patent has been filed with the Indian Patent Office (IPO) (Appendix B).

(2) A standardised protocol calibrated to the Indian Railways stress window (275–523 kPa), recording per-blow peak force and five-milestone settlement—a full force-and-deformation history, versus the single post-test fines percentage of conventional Aggregate Impact Value testing.

(3) A morphological-analysis method (Appendix B) that ingests paired pre/post STL meshes and exports twenty-two descriptors per particle to a multi-sheet Excel workbook for direct comparison.

6.2.2 Scientific findings

(1) Empirical demonstration that base condition (sand cushion vs. rigid concrete) governs the selection between Mode A (diffuse abrasion) and Mode B (catastrophic fracture) under drop-weight impact loading of ballast, using a paired per-particle 3D morphological method. All six observed Mode B events occurred on the rigid base.

(2) A heat-map of mean per-particle %-change across fourteen descriptors for nine configurations: sign-uniformity in size and primary-shape descriptors, heterogeneity in shape-ratio descriptors, and sphericity and convexity as more robust diagnostics than angularity and roundness.

(3) Empirical demonstration that pad performance is energy- and base-dependent: PUBM dominates at the FB-300 and RB-450 corners; RUBM edges ahead on RB-300. A single optimal pad cannot be specified independent of the substructure and loading energy.

(4) The first published nine-configuration factorial dataset pairing large-diameter drop-weight impact loading (RDSO gradation, Indian Railways stress levels) with per-particle 3D morphology—a reference for validating discrete-element models and machine-learning degradation predictors.

6.3 Limitations of the study

Four limitations bound the generalisability of the conclusions and should be recorded in any reading of the findings.

Limitation 1 – Sample size per configuration

Five tracer particles supports the pooled Mode A significance test and Mode B detection, but not per-configuration ANOVA or tight Mode B probability estimates. Expanding to 10–20 tracers is the single most effective follow-up improvement.

Limitation 2 – Single parent rock type

Only basalt from one Vadodara quarry was tested. Parent rock affects both Mode A abrasion rate and Mode B fracture threshold, so generalisation requires extension to another RDSO-approved rock.

Limitation 3 – Surface morphology only

The EinScan structured-light scanner cannot detect sub-surface micro-cracks, incipient fracture planes, or mineralogical alteration. Mode A particles may carry latent damage that accelerates Mode B fracture; X-ray CT would be required to detect this.

Limitation 4 – Laboratory scale

The study is limited to a single mould diameter and a single hammer mass. Field-scale validation, including the effect of variable train speeds and long-term cyclic loading, is outside the present scope.

6.4 Recommendations for future work

Five concrete directions for follow-up research are identified, in rough order of increasing scope. Each would usefully build on the methodological platform established by the present thesis.

(1) **Expanded tracer-count study.** Repeat the present nine-configuration programme with 15–20 tracer particles per configuration to support per-configuration ANOVA and tighter estimates of Mode B event probabilities.

(2) **Multi-rock-type extension.** Extend the programme to granite and quartzite ballast, using the same nine-configuration matrix and the same per-particle morphology pipeline, to map the parent-rock dependence of the Mode A / Mode B crossover.

(3) **AI/ML-based prediction of degradation behaviour.** Use the per-particle paired morphological dataset to train supervised machine-learning models. A trained surrogate would allow rapid screening of pad–base combinations without running the full impact campaign, and feature-importance analysis would identify which morphological descriptors most strongly govern degradation outcomes.

(4) **Cyclic-plus-impact combined loading.** Develop a composite loading method that applies cyclic-triaxial loading (mimicking ordinary traffic) interrupted periodically by drop-weight impact events (mimicking transition-zone hot-spots), and report the corresponding Mode A / Mode B evolution. This is the closest laboratory analogue to real service loading.

(5) **Field correlation.** Couple the laboratory degradation metrics with in-situ measurements at instrumented bridge approaches, rail joints, or transition zones on Indian Railways, to calibrate the laboratory predictions against observed track-geometry degradation rates.

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APPENDIX A

Tabulated Data Excerpts

This appendix reproduces tabulated excerpts from the principal laboratory data files referenced throughout the thesis. The full raw data—including all 45 paired before–after scans and 22 descriptors per mesh—is archived in the two Excel workbooks referenced in Appendix B and is available from the authors on reasonable request.

A.1 Test matrix — full specification

Table A.1. Complete nine-configuration test matrix with measured peak force, kinetic energy, and the legacy → systematic test-number mapping. Specimen mass is approximately 20 kg per test; ballast-column height is approximately 280 mm at the 1200 kg/m^3 reference density.

Test	Systematic label	Legacy label	Pad	Base	h (mm)	K_e (J)	Peak F (kN)	σ (kPa)
T1	PUBM–FB–300	PUBM-FB-LH	PUBM	FB	300	147	19.4	275
T2	RUBM–FB–300	RUBM-FB-LH	RUBM	FB	300	147	19.4	275
T3	UR–FB–300	UR-FB-LH	UR	FB	300	147	19.4	275
T4	UR–RB–300	UR-RB-LH	UR	RB	300	147	19.4	275
T5	RUBM–RB–450	RUBM-RB-HH	RUBM	RB	450	221	37.0	523
T6	PUBM–RB–450	PUBM-RB-HH	PUBM	RB	450	221	37.0	523
T7	UR–RB–450	UR-RB-HH	UR	RB	450	221	37.0	523
T8	RUBM–RB–300	RUBM-RB-LH	RUBM	RB	300	147	19.4	275
T9	PUBM–RB–300	PUBM-RB-LH	PUBM	RB	300	147	19.4	275

A.2 Per-particle paired morphology — summary table

Table A.2. Per-configuration Mode A summary of the principal morphological-change descriptors. Mode B outliers excluded. ΔV, $\Delta\Psi$, ΔC in %-relative change. n = Mode A sample size in each configuration.

Test	Label	n (Mode A)	Mode B events	Mean $\%\Delta V$	Mean $\%\Delta\Psi$	Mean $\%\Delta C$
T1	PUBM–FB–300	5	0	−0.18	+0.56	+0.50
T2	RUBM–FB–300	5	0	−0.10	+0.25	+0.29
T3	UR–FB–300	5	0	−0.44	+0.65	+0.24
T4	UR–RB–300	3	2	−0.87	+1.43	+1.31
T5	RUBM–RB–450	4	1	−0.84	+0.31	+0.27
T6	PUBM–RB–450	4	1	−0.30	+0.50	+0.75
T7	UR–RB–450	4	1	−0.72	+0.55	+0.18
T8	RUBM–RB–300	4	1	−0.58	+0.68	−0.01
T9	PUBM–RB–300	5	0	−0.18	+0.48	+0.26
All (pooled)		**39**	**6**	**−0.47**	**+0.56**	**+0.40**

A.3 Complete post-test sieve data

Table A.3. Percentage retained on each sieve, post-test, for the nine configurations. Rows are sieve sizes; columns are tests. Values in weight percent of the post-test specimen mass.

Sieve (mm)	T1	T2	T3	T4	T5	T6	T7	T8	T9
	PUBM-FB-300	RUBM-FB-300	UR-FB-300	UR-RB-300	RUBM-RB-450	PUBM-RB-450	UR-RB-450	RUBM-RB-300	PUBM-RB-300
63	7.18	8.34	8.44	12.11	8.01	6.45	8.35	11.00	8.63
40	9.15	8.09	6.44	12.25	12.15	10.85	9.62	13.02	9.26
31.5	2.11	2.65	2.15	2.44	1.55	5.75	1.50	3.44	2.53
20	0.210	0.375	0.500	0.675	0.270	0.200	0.125	0.250	0.270
16	0.010	0.110	0.050	0.090	0.036	0.100	0.080	0.010	0.033
12.5	0.004	0.043	0.037	0.120	0.021	0.015	0.150	0.022	0.052
10	0.007	0.026	0.030	0.070	0.030	0.011	0.065	0.010	0.028
4.75	0.012	0.054	0.044	0.130	0.071	0.021	0.130	0.041	0.040
2.36	0.015	0.009	0.019	0.038	0.032	0.009	0.047	0.019	0.016
1.18	0.010	0.009	0.014	0.026	0.025	0.006	0.046	0.014	0.012
0.6	0.025	0.005	0.025	0.014	0.015	0.004	0.037	0.010	0.006
0.075	0.022	0.013	0.022	0.035	0.050	0.011	0.083	0.040	0.024
Pan	0.005	0.004	0.005	0.013	0.011	0.004	0.024	0.010	0.004
Specimen Total (g)	18.76	19.73	17.78	28.01	22.27	23.43	20.26	27.89	20.91
$P_{0.075}$ (%)	0.12	0.07	0.12	0.12	0.22	0.05	0.41	0.14	0.11
P_{20} (%)	0.50	1.09	0.95	1.16	1.01	0.26	2.28	0.62	0.77
FI_{IN} (%)	0.62	1.16	1.07	1.28	1.23	0.31	2.69	0.76	0.88

APPENDIX B

Python Morphological Analysis Script

This appendix records the Python morphological-analysis script. The script reads STL meshes from a pre-test folder and a post-test folder, performs the four-check mesh quality-control gate, computes the twenty-two morphological descriptors, pairs the pre-test and post-test meshes by filename, computes the per-particle changes, and exports the complete dataset to a multi-sheet Excel workbook.

B.1 Software environment

- Python 3.10 or later (tested on 3.12).
- `trimesh` $\geq$ 4.0 — STL input, mesh integrity checks, convex hull (Dawson-Haggerty, 2019).
- `numpy` $\geq$ 1.24 — numerical operations on descriptor arrays (Harris et al., 2020).
- `pandas` $\geq$ 2.0 — DataFrame handling for the paired dataset.
- `openpyxl` — Excel workbook output.
- `scipy` — linear algebra and statistics (Virtanen et al., 2020).
- `numpy-stl` — legacy STL-format compatibility layer.

Installation is via the pip toolchain:

```
pip install trimesh numpy pandas openpyxl scipy numpy-stl
```

B.2 Pipeline script — `ballast_analyzer.py`

The complete pipeline script is archived as `ballast_analyzer.py` (approximately 450 lines of commented Python). The core descriptor-extraction loop is reproduced in Listing B.1; the full source is available from the authors.

```
import os, trimesh, numpy as np, pandas as pd

# ---------- Four-check quality gate ----------
def mesh_ok(mesh):
    return (mesh.is_watertight
            and mesh.body_count == 1
            and mesh.volume > 10_000
```

```python
            and mesh.volume < 250_000)

# ---------- Descriptor extraction ----------
def extract_descriptors(stl_path):
    m = trimesh.load(stl_path)
    if not mesh_ok(m):
        raise ValueError(f"{stl_path}: mesh QC failed")

    V, A = m.volume, m.area
    ch = m.convex_hull
    V_ch, A_ch = ch.volume, ch.area

    # Principal dimensions from oriented bounding box
    obb = m.bounding_box_oriented.extents          # [S, I, L]
    S, I, L = sorted(obb)

    # Sphericity (Wadell, 1932)
    Psi = (np.pi ** (1/3)) * ((6 * V) ** (2/3)) / A

    # Convexity
    C = V / V_ch

    # Form descriptors
    F, E, AR = S/I, I/L, L/S
    FI = (L + I) / (2 * S)

    # Surface texture
    T = A / A_ch
    D_eq = (6 * V / np.pi) ** (1/3)

    return dict(V=V, A=A, D_eq=D_eq, S=S, I=I, L=L,
                V_ch=V_ch, A_ch=A_ch, Psi=Psi, C=C,
                F=F, E=E, AR=AR, FI=FI, T=T)

# ---------- Pair before & after, write workbook ----------
def run(before_dir, after_dir, out_xlsx):
    rows_b, rows_a = [], []
    for fn in sorted(os.listdir(before_dir)):
        if not fn.endswith(".stl"):
            continue
        d_b = extract_descriptors(os.path.join(before_dir, fn))
        d_a = extract_descriptors(os.path.join(after_dir,  fn))
        d_b["Particle"] = fn.replace(".stl", "")
        d_a["Particle"] = fn.replace(".stl", "")
        rows_b.append(d_b)
        rows_a.append(d_a)
```

```

    df_b = pd.DataFrame(rows_b).set_index("Particle")
    df_a = pd.DataFrame(rows_a).set_index("Particle")
    df_delta = df_a - df_b
    df_pct   = 100 * df_delta / df_b

    with pd.ExcelWriter(out_xlsx) as xw:
        df_b.to_excel(xw,     "Before Test")
        df_a.to_excel(xw,     "After Test")
        df_delta.to_excel(xw, "Changes (After-Before)")
        df_pct.to_excel(xw,   "Percent Changes")
```

Listing B.1. Core extraction logic of `ballast_analyzer.py`.

APPENDIX C

Statistical Methods

All statistical inference is a conversation between the data and the assumptions.

— after George Box

In this chapter

This appendix gives formal definitions for the three statistical procedures used in Chapter 5: the Shapiro–Wilk test for normality, the Wilcoxon signed-rank test for paired changes when the normality assumption fails, and the Tukey $1.5\times$IQR criterion for outlier identification on the volume-change distribution. The appendix also documents the significance thresholds, the multiple-comparison correction, and the power-analysis rationale for the chosen sample size ($n = 45$ tracer particles).

C.1 Notation

In this appendix, $\{x_i\}_{i=1}^{n}$ denotes a sample of size n drawn from a population with unknown distribution. For paired data (pre-test / post-test of the same particle), $\{d_i\}_{i=1}^{n}$ denotes the per-particle change, $d_i = x_i^{\text{after}} - x_i^{\text{before}}$. The median of a sample is denoted $\tilde{x}$ and the order statistics by $x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)}$.

C.2 Shapiro–Wilk test for normality

The Shapiro–Wilk statistic W tests the null hypothesis H_0 that a sample is drawn from a normal distribution against the alternative H_1 that it is not. The statistic is

$$W = \frac{\left(\sum_{i=1}^{n} a_i\, x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \tag{C.1}$$

where $\bar{x}$ is the sample mean and the coefficients $\{a_i\}$ are tabulated constants derived from the expected values and the covariance matrix of the order statistics of a standard normal sample. The statistic lies in the closed interval $[0, 1]$; values close to 1 favour H_0, and small values reject it.

Application in this thesis. The test is applied in Section 5.1 to the paired volume-change distribution $\{\%\Delta V_i\}$ across the $n = 45$ tracer particles. A rejection of H_0 ($p < 0.05$) motivates the use of a non-parametric paired test (Wilcoxon signed-rank) in place of the parametric paired t-test.

C.3 Wilcoxon signed-rank test for paired data

The Wilcoxon signed-rank statistic T^+ tests the null hypothesis H_0 that the median of the paired-difference distribution is zero against the alternative H_1 that it is non-zero, without requiring the paired differences to be normally distributed. The procedure is:

1. Compute the per-pair differences $d_i = x_i^{\text{after}} - x_i^{\text{before}}$.
2. Discard any pairs with $d_i = 0$ and reduce n accordingly.
3. Rank the absolute values $|d_i|$ from 1 (smallest) to n (largest), assigning the mean rank to ties.
4. Compute
$$T^+ = \sum_{i:\, d_i>0} R_i, \qquad T^- = \sum_{i:\, d_i<0} R_i, \tag{C.2}$$
where R_i is the rank of $|d_i|$.
5. The smaller of T^+ and T^- is referred to the Wilcoxon-signed-rank reference distribution to obtain a two-sided p-value; for $n > 20$ the normal-approximation z-statistic
$$z = \frac{T^+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}} \tag{C.3}$$
is used.

Application in this thesis. The test is applied to the paired sphericity, convexity, angularity, roundness, and volume changes across all $n = 45$ tracer particles. The sphericity result reported in Section 5.2 is $z = -5.84$, $p < 10^{-6}$, supporting the universal Mode A (diffuse-abrasion) claim with overwhelming statistical evidence.

C.4 Tukey 1.5×IQR outlier criterion

For a sample with inter-quartile range $\mathrm{IQR} = Q_3 - Q_1$, the Tukey outlier fence places a point x_i in the outlier set if either

$$x_i < Q_1 - 1.5 \cdot \mathrm{IQR} \qquad \text{or} \qquad x_i > Q_3 + 1.5 \cdot \mathrm{IQR}, \tag{C.4}$$

where Q_1 and Q_3 are the first and third quartiles of the sample respectively. The factor 1.5 is the classical Tukey choice; under a normal distribution, this fence excludes approximately 0.7 % of the population on each tail. The criterion is robust because Q_1, Q_3, and IQR are themselves robust location-and-scale estimators.

Application in this thesis. The criterion is applied to the paired volume-change distribution $\{\%\Delta V_i\}$ in Section 5.1. The Q_1, Q_3 values are computed from all $n = 45$ paired records; particles falling below $Q_1 - 1.5\,\mathrm{IQR}$ are labelled Mode B (catastrophic-fracture candidates). The six identified outliers are exactly the six particles of Figure 5.1.

C.5 Significance thresholds and multiple-comparison correction

The significance threshold throughout the thesis is the conventional $\alpha = 0.05$ unless stated otherwise. Where a family of k comparisons is made simultaneously (the sphericity, convexity, angularity, and roundness Wilcoxon tests of Section 5.2, $k = 4$), the Bonferroni-corrected threshold $\alpha/k = 0.0125$ is applied to control the family-wise Type-I error rate. All four descriptors meet this stricter threshold; the principal sphericity result ($p < 10^{-6}$) is well below it.

C.6 Sample-size rationale

The sample size of $n = 45$ tracer particles (five per configuration across nine configurations) was determined *a priori* from a two-sided paired-test power analysis. With significance level $\alpha = 0.05$ and target power $1 - \beta = 0.80$, a paired test on the volume-change distribution can detect an effect size of $|d|/\sigma_d \approx 0.42$ at $n = 45$. The smallest practically meaningful effect size in the morphological literature is $|d|/\sigma_d \approx 0.5$ (corresponding to a sphericity change of $\Delta\Psi \approx 0.003$ against a typical within-population standard deviation of $\sigma_\Psi \approx 0.006$). The chosen sample size therefore comfortably exceeds the power requirement for the principal effect of interest.

Reproducibility

All statistical procedures in this appendix are implemented in the `scipy.stats` module of the Python pipeline of Appendix B: `scipy.stats.shapiro`, `scipy.stats.wilcoxon` (with `zero_method="wilcox"` and `correction=True`), and a hand-coded Tukey fence based on `numpy.percentile`. The pipeline produces deterministic output from the paired STL meshes and is therefore fully reproducible by any reviewer with access to the raw data.

Chapter takeaways

- **Shapiro–Wilk** verifies that the parametric-normality assumption fails for the paired volume-change data; the non-parametric Wilcoxon test is therefore the correct inference tool.
- **Wilcoxon signed-rank** confirms that Mode A (diffuse abrasion) is universal: the median paired sphericity change is significantly greater than zero, $p < 10^{-6}$.
- **Tukey** $1.5\times$**IQR** identifies six Mode B (catastrophic-fracture) outliers in the volume-change distribution; all six are drawn from rigid-base configurations only.
- All three procedures are implemented in the open Python pipeline of Appendix B.

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Gati Shakti Vishwavidyalaya, Vadodara
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— End of thesis —